

E-JETS FAMILY

2.0

USER MANUAL

X-CRAFTS

DISCLAIMER

The information contained in this **document is for simulation use only**, within the X-Plane flight simulator. This document may not always be up to date with the latest version of the E-Jets Family product. This document is intended for entertainment only, and may not be used in situations involving real-life aircraft, or real-life aviation.

This document may be copied and distributed by X-Crafts customers for entertainment only.

FMS DOCUMENTATION

All FMS related documentation can be found on our [website](#). We plan to create a separate FMS manual in a future update. In the meantime, you can learn how to use the FMS by following the [Demo Flight instructions](#) on our website.

On the Demo Flight page, we provide detailed instructions to fully load and edit a flight plan, do a DIRECT TO and other operations. We walk you through the entire process in our Demo Flight video. We also explain how to use the FMS in conjunction with the EFB, and how to enter the performance data.

DEMO FLIGHT

This is the best place to start. You can follow simple and complete instructions with videos to learn how to start the plane, program the FMS, use the EFB, and complete a full flight starting in from Bergamo in Italy, climbing over the Alps to then land in the Austrian capital Vienna.

[Link to our Demo Flight](#)

Once you have completed the demo flight, you can use this manual for a deep dive into all the systems of the aircraft.

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INTRODUCTION

Hello and thank you for purchasing one or more of the E-Jets Family products!

We hope you have as much enjoyment flying your new aircraft as we did creating it and improving it to its latest version.

This manual is applicable to all of the E-Jets Family by X-Crafts products: E170, E175, E190, E195, and Lineage 1000.

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TECH SUPPORT

If you need any customer support, please feel free to visit us at:

www.xcrafts.com/tech-support

GETTING STARTED

TUTORIALS & DEMO FLIGHTS

Learning how to fly complex airliners can sometimes be frustrating and time consuming. We have designed a Getting Started page which will make this process fun and quick.

This is the one place where you will find all the Demo flights with simple instructions and videos, detailed FAQs, Tutorials and more. Additionally, we are always here for you in our forums if you need any help getting started.

All can be found here: <https://www.xcrafts.com/ejets-family/getting-started>

INSTALLATION

To install any of the aircraft or the bundled set(s), simply extract the zipped folder(s) into X-Plane/Aircraft/... in a folder of your choice.

- We suggest that the folder name and path contain no non-unicode characters.
- The folder also must not be a system folder.
- If you have more than one X-Crafts product, you can have them all in one *X-Crafts* sub-folder like this (*X-Plane/aircraft/X-Crafts*).

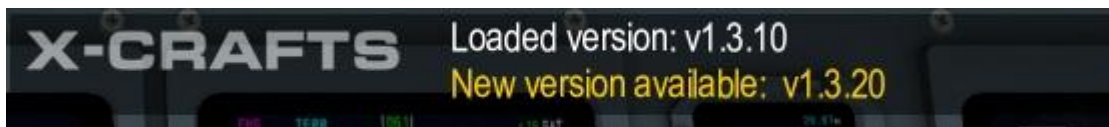
ACTIVATION

You will need to activate your products before you can use them. Please follow the activation instructions on our website: <https://www.xcrafts.com/activate>

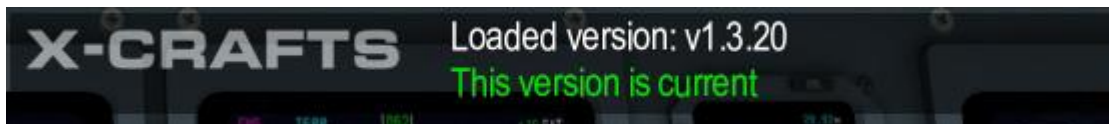
VERSION CHECKING

- The ERJ Family uses the SkunkCrafts updater to enable pilots to keep their product current. This is covered on our website www.xcrafts.com/update-download
- Every time you load the plane, a pop-up will appear in the lower right corner informing you whether you are using the most up to date version.
- This is done with a very brief check of the online update repository and shares no information of any kind whatsoever.

The pop-up will look like this if an update is available:



The pop-up will look like this if you are up to date:



You will also see other messages if for some reason the version checker runs into issues. This can be anything from a loss of internet connectivity to some kind of local file error in the updater configuration.

This message appears during aircraft initialization and then fades out. You can dismiss it immediately simply by clicking on it. This last bit may be especially useful if you immediately start clicking on things the second you get into the cockpit and the update notification inhibits your mouse focus.

VIEWING OPTIONS

FIELD OF VIEW

Recommended field of view is 70-80 degrees. Change this in X-Plane under Settings > Rendering Options > Special Viewing Options > Lateral field of view

QUICK LOOKS

For ease of viewing different parts of the cockpit, cabin or exterior, you can set a custom Quicklook . The steps are as follows:

1. In the 3D cockpit view, set the desired view. Or even an exterior view ...
2. Now press Ctrl+ "1" (or any other digit from 0 to 9). The number must be on the number pad on the keyboard.
3. Thus stored, you now can see this view by pressing the "1" (without ctrl) on the number pad.
4. Pressing "1" again, the previous view is restored.
5. These 20 Quick Looks can be tied to different joystick buttons. This can be done as usual in the menu "joystick, keyboard and devices" - "Extra Buttons".
6. The saved views are aircraft specific. They are saved directly in the appropriate directory of the Aircraft, stored as "Aircraft Name_prefs.txt"

X-Plane offers up to 20 Quick Look's, but you have to assign the storage commands to keys other than the 10 original storage (CTRL + ??) on the number pad.

POP-UP SCREENS

The following screens can be viewed in a pop-up view, or a pop-out view: PFD, MFD, EICAS, FMS, EFB and the Lineage 1000 Light Control Panel. This gives you the ability to open them in a 2D view and move them anywhere on your screen, or even open them in a separate window on your computer. This is particularly useful if you want to permanently draw them on a separate monitor.



2D Pop-Up Controls

1. Opening Pop-Ups:

- You can simply left-click on any of the screens to open them.
- Open them from the Top X-Plane bar X-Crafts Dropdown menu.
- Assign commands to toggle pop-ups:
 - "xcrafts/pfd_panel_toggle"
 - "xcrafts/mfd_panel_toggle"
 - "xcrafts/eicas_panel_toggle"
- Find these commands in keyboard settings under "xcrafts" (note the lowercase).

2. **Closing and Dragging:**

- Right-click anywhere outside interactive spots to close the pop-up.
- Hold the left mouse button to drag the window.

3. **Resizing:**

- Use the mouse wheel anywhere outside the interactive spots.
- Use the resize area at the bottom right corner (a red triangle will appear when the cursor is in the zone).
- Resizing maintains the window's aspect ratio.

4. **Window Options:**

- Icons for closing, popping to system, and locking appear at the top when the cursor is in the zone.
- Mouse wheel resize, right-click close, and position lock won't work if the windows are popped out into a separate window.

PFD and MFD Interactivity

Some of the screens also have interactive spots, which allow you to change certain values directly on the screens, instead of using the glareshield panel.

1. **Click Spots:**

- Enable "Show Instruments Click Regions" under the View menu in X-Plane to see the click spots.
- Operate click spots by clicking or using the mouse wheel.
- Arrows indicate the direction of change for click actions.

2. **PFD Controls:**

- Change any value using standard controls.
- Swap Radio and Nav frequencies by clicking on NAV1 or COM1.
- Tune frequencies using zones over the numerical values.
- Toggle AP and AT by clicking their zones.
- Adjust Autopilot Airspeed and Altitude:
 - Altitude set zone is divided into thousands (left) and hundreds (right).
- Visible zones for adjustable values (e.g., QNH, CRS, HDG, VVI, MINIMUMS).

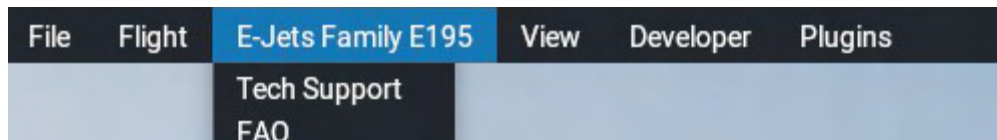
3. **MFD Controls:**

- Works the same as the 3D MFD.
- Cursor changes to a cross like in the real plane.
- Terrain Radar plugin does not render on the 2D pop-up (no fix available at the moment).

PLUGINS & SETTINGS

The product comes with several plugins and settings menus.

X-PLANE TOP BAR DROP DOWN MENU



This dropdown menu gives you access to:

- Direct link to our Tech Support
- Direct link to our FAQ page
- Direct access to the Screen Pop Ups

EFB

The **EFB (Electronic Flight Bag)** is our in-house plugin that helps flight crews perform flight management tasks more easily and efficiently.

- We have created a comprehensive video tutorial where we cover all features of the EFB and how to use them.
- The information in this manual mainly describes the features included, but we advise you to watch the tutorial to learn how to use each feature properly
- You can watch it [here](#).



OPENING THE EFB IN A POP-UP WINDOW

1. Done by clicking on the **“Home” Button** of the tablet.
2. Through the X-Plane top bar **Drop-down menu**
3. By assigning a **command to a key** on your keyboard.
 - You will find the command under X-Plane Settings > Keyboard > Search for **XCrafts** > **“efb1”**.

Note: Sometimes, when trying to close the Pop-Up EFB by the keyboard command, there is a conflict of keystrokes, where the EFB won't know if you are trying to type into the EFB, or you are trying to close the EFB. In such case, we recommend that you, either:

- A. Close the EFB with your mouse, by clicking on the red button in the top left corner of the window.
- B. Click somewhere outside the EFB screen to disable the EFB key entry focus, and then press the key on your keyboard to close it.

HIDING THE EFB

- You can hide the EFB by clicking on the clipboard below the EFB.

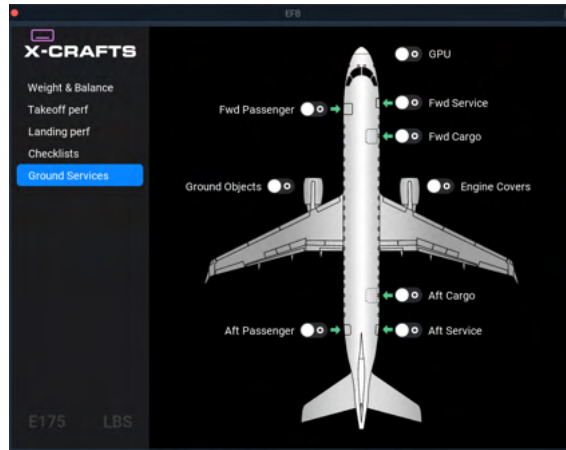


ROTATING THE EFB

- You can rotate and reposition the EFB by grabbing onto its sides.

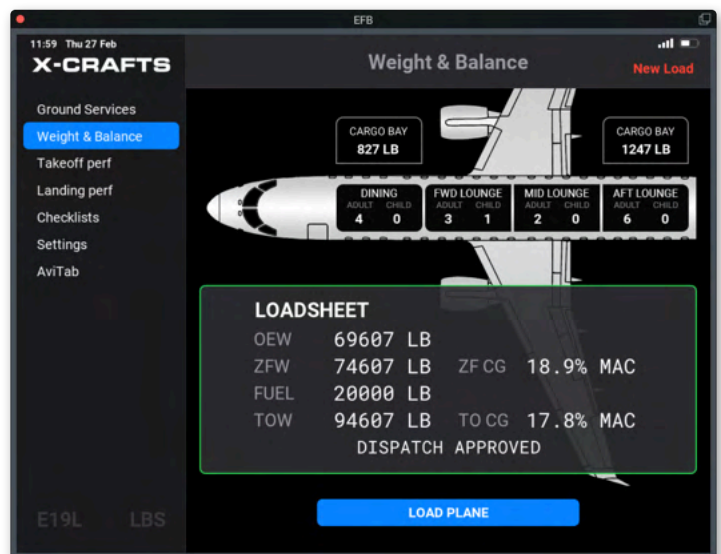
GROUND SERVICES

- Opening and closing of the doors.
- Connecting the GPU.
- Toggling the Chocks, Remove before flight tags, and Engine Covers.



WEIGHT & BALANCE

- To get a weight and balance Loadheet, you have to enter the Payload or Zero Fuel Weight, and Fuel Weight.
- You can enter these manually or fetch them directly from your last generated Simbrief flight plan by entering your Simbrief Pilot ID as shown in the tutorial.
- The number in each of the zones is adults and (children).



- You can accept the load sheet by clicking on **LOAD PLANE**, which will load the plane according to the load sheet.

TAKEOFF & LANDING PERF

- These pages can be used to calculate Takeoff & Landing Performance data and send it to the FMS
 1. Fill out the input fields > Press **COMPUTE**.
 2. You can then enter the Vspeeds manually into the FMS, or you can send the VSpeeds to the FMS by pressing "**Send to FMS**" in the top right corner.
 - You can then load the vspeeds in the **FMS**: Press the **DLK** (datalink) button. You will now see the landing speeds option highlighted green. If you press the right soft key next to the highlighted text, you will load the vspeeds
- You can enter HPA or IN values for QNH, and the tablet will use the correct units based on your format
- You can use the same process for landing speeds from the Landing perf page on the EFB.

EFB

X-CRAFTS

Performance - Takeoff [Send to FMS](#)

Weight & Balance
Takeoff perf
Landing perf
Checklists
Ground Services

AIRPORT LOWW 79580 LBS T/O WEIGHT

RUNWAY 11 TO-1 THRUST

CONDITION DRY F1 FLAPS

WIND 245/8

TEMP 15 °C

QNH 1023 HPA

COMPUTE

FLAPS F1 V1 142

RWY 11 VR 152

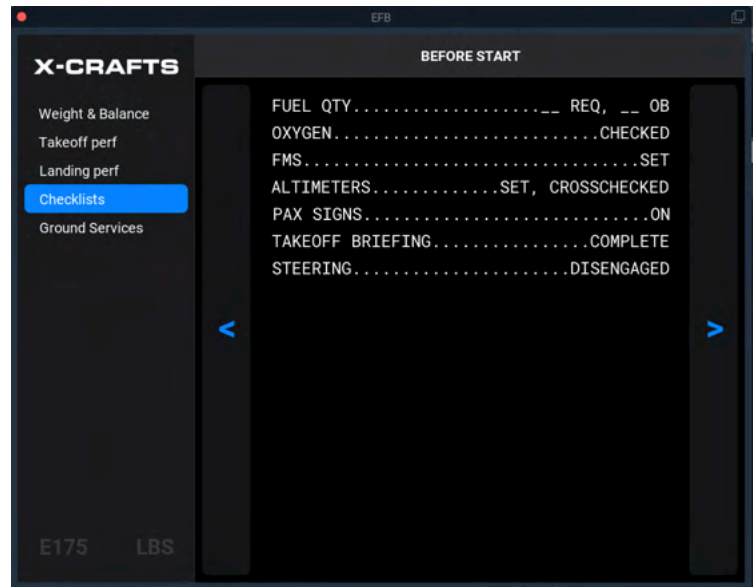
TOGW 79580 LBS V2 153

VFS 200

E175 LBS

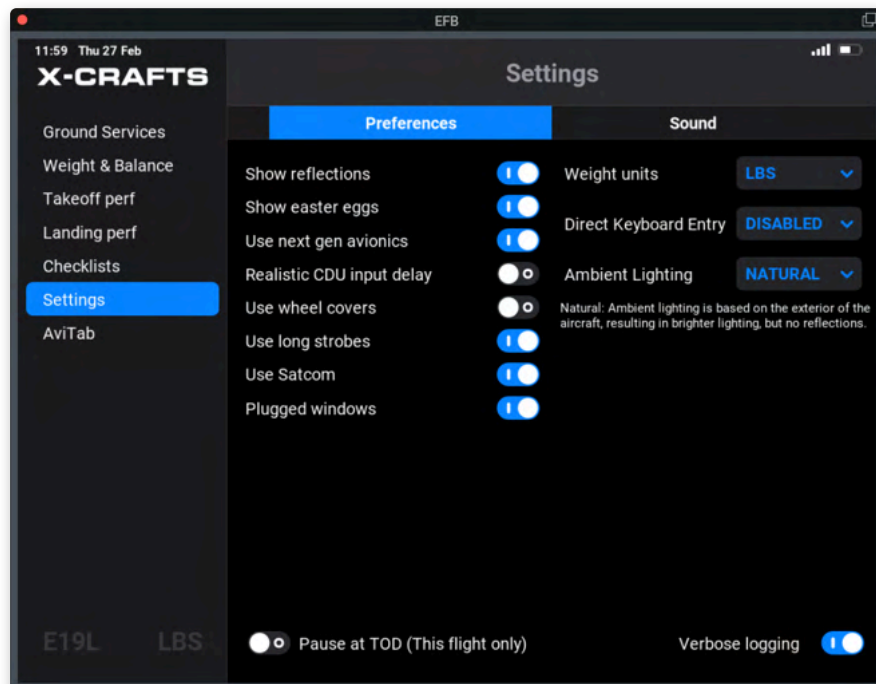
CHECKLISTS

- You can access the checklists and you can also modify it based on your preference.
- Watch our video at the beginning of this section for detailed explanation.



SETTINGS MENU

This tab intended for setting up default preferences, as it will remember the settings between sessions.



PREFERENCES

Allows you to set various preferences for your flight.

Show reflections

- Toggles the screen reflections

Show Easter eggs

- Toggles the Easter eggs.

Use next gen avionics

- This toggles between the older Load 25 and Load 27 visuals for the PFD.
- In real life, there are many differences across all the avionics between these two versions, however, we have developed everything based on Load 27, and the only change is the visual representation of the PFD.

Realistic CDU input delay

- The real FMS has a bit of a delay when inserting data, pressing the side soft keys or changing pages. We have added this delay to emulate the real feel.
- You can disable this delay, in which case the pages will respond instantly

Use wheel covers

- Toggles the wheel covers

Use Long LED Strobe/Beacon lights

- Some E-Jets are equipped with newer LED lights with a different flashing frequency.

Use Extended winglets

- Toggles between the standard and wide winglets
- Also changes the strobe and beacon lights
- Only available on the E175

Use SATCOM

- Enable the SATCOM antenna on top of the fuselage

Use Plugged windows

- Only available on the Lineage 1000
- Hide some windows based on the Lineage 1000 fuselage

Weight Units

- Set your preferred units metric/imperial

Direct Keyboard Entry (DKE) Key

- When DKE is active, you can use your keyboard to type directly into the scratchpads of the MCDU. The default key for triggering DKE is <TAB> on your keyboard, however, this key is sometimes used in other applications or you may have it assigned to any other function.
- This setting allows you to change or disable the default toggle for DKE to be <CAPS LOCK>

Ambient Lighting Modifier

- We're introducing an unprecedented **Ambient Lighting Modifier** with three options:
 - **Default** (X-Plane's lighting)
 - **Dynamic** (Lighting adjusts as you move through the cabin)
 - **Natural** (Lighting based on the aircraft's exterior—this makes the interior much brighter!)

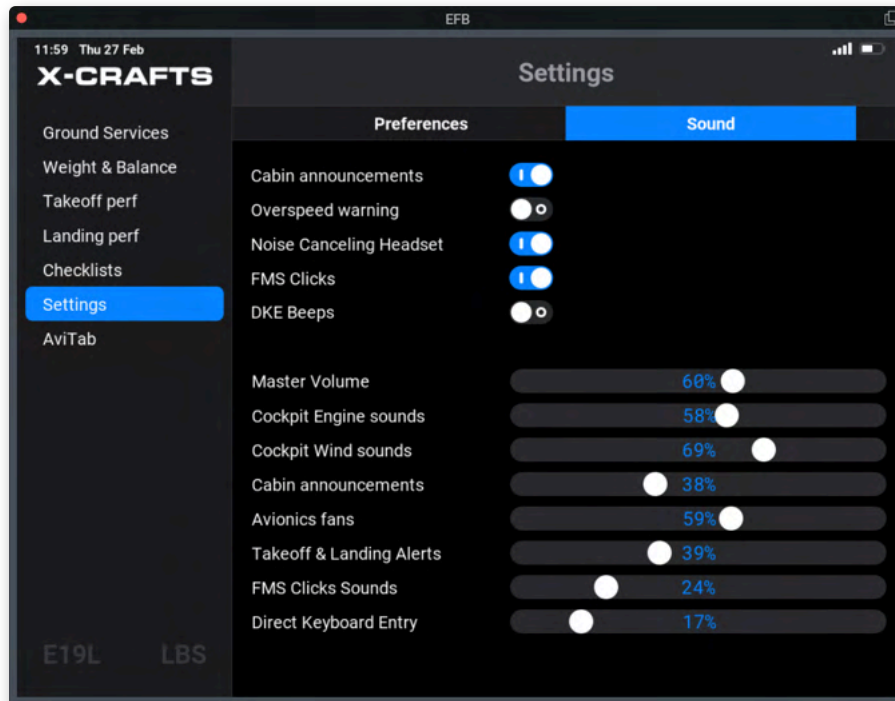
Pause at Top of Descent

- The simulator will automatically pause before it reaches Top of Descent. This is useful if you want to step away from the computer while in cruise.

Enable verbose logging

- If there is any technical issue, we may ask you to enable this so that more information is written out into the log.txt. This would help us better diagnose your issue.

SOUND SETTINGS



- Allows you to set various sound preferences.
- We recommend to keep all sliders at 50% for the most realistic experience
- DKE Beeps will toggle the beeping that your keyboard produces when you have direct Keyboard Entry enabled

AVITAB

- Avitab is a third-party plugin that is useful for a very wide variety of functions. It can be downloaded for free on [X-Plane.org](https://www.x-plane.org) from [this link](#). Support is also offered in [this thread](#).
- If Avitab is installed, a new TAB called Avitab will display on the EFB below Ground Services. This will allow you to access Avitab via the EFB.
- You can open Avitab in a separate Pop-Up window by clicking on the tablet “Home” button, same as the EFB.
- **Note:** Any issues related to Avitab are out of our hands, we are simply providing Avitab as it is, but if it doesn't work, that would need to be reported to the developers of it.



USING YOUR KEYBOARD FOR DIRECT INPUT FOR THE FMS, EFB & AVITAB

DIRECT KEYBOARD ENTRY (DKE)

- DKE function is only used for the **FMS**. If DKE is active, you will not be able to use the EFB or Avitab.
- Toggling DKE is done by:
 - a. Pressing TAB on your keyboard (or CAPS LOCK, based on your preference in the Settings menu.)
 - b. Clicking on the MDCU scratchpad
- If DKE is active, you'll hear a beep for each keystroke. This is designed so that you are aware that DKE is enabled.
- You can adjust the volume of the beep in the Settings menu.

EFB

- Clicking on the screen enables the focus on the screen and allows you to enter any input with your keyboard. The focus is indicated by a blue edge around the screen.
- If you click outside of the EFB screen, the blue edge will disappear, which means the focus is no longer on the EFB screen and you can use your keyboard as usual.
- In the Pop-Up EFB window, input focus is indicated by a magenta blinking keyboard icon

Note: If the screen edge of the EFB is blue, **but the EFB is not registering any keyboard input**, try using the pop-up version of the EFB where there should be no issue with the keyboard input.

AVITAB

- Clicking on the screen enables the focus on the screen and allows you to enter any input with your keyboard.
- You can also bring Avitab up in the 2D Popup window, by clicking on the "Tablet Home button"

RECOMMENDED 3RD PARTY PLUGINS

SKUNKCRAFTS UPDATER

We recommend that you use the SkunkCrafts Updater which will automatically detect any new updates, and will allow you to install them through the plugin interface, bypassing the need to redownload the aircraft from the store after every update.

[You can get the updater here.](#)

More information about how to use the Skunkcrafts Updater can be found on [our website](#).

TERRAIN RADAR

We recommend that you download and install the Terrain Radar plugin by DrGluck. This plugin enables the MFD to display Terrain in MAP mode.

[You can get the Terrain Radar here.](#)

AIRCRAFT OPERATION

AUTOPILOT & TAKEOFF MODES

AUTOPILOT

- Autopilot engagement:
 - AP button is pushed on the Guidance panel
 - AP/FD button is pushed on the yoke.
- Autopilot disengagement:
 - Either AP/TRIM DISC button is pushed on the yoke.
 - AP button is pushed on the Guidance panel.
 - Any manual pitch trim is activated.
 - Stick Shaker activated.
- Default modes for the autopilot are: Roll Hold (ROLL) and FPA.
 - **Note:** In the current version of the product, The default vertical mode is Pitch Hold, which is essentially FPA, except we do not have the ability to change the FPA. So if you want to change the FPA angle, just hit FPA on the guidance panel again.

TAKEOFF MODES

There are several Take off settings and modes related to either the, Thrust Rating Setting (TRS), Autothrottle or Lateral and Vertical guidance. Understanding the differences is key to properly configuring the aircraft for takeoff.

1. THRUST RATING SETTING (TRS) TAKEOFF MODES

- Unrelated to lateral and vertical guidance.
- TRS limits the N1 for Takeoff based on the setting in the MCDU.
- Represented on the EICAS engine section as **TO-1** or **TO-2**, accompanied with the ATCS setting.
- More detailed information on the TRS can be found in the FMS manual.

2. AUTOTHROTTLE TAKEOFF MODE (TO AT)

- Represented on the **left** side of the FMA.
- Engaged by pushing the TO/GA button on the throttle levers when on ground.
- Can be disengaged by pulling the throttles back to idle and pressing AT on the guidance panel.

Arming TO AT mode:

1. TLA (Throttle Lever Angle) must be below the TO/GA position.
 2. AT on the guidance panel must be pressed.
- TO AT arming is confirmed by a white **TO** on the FMA.

Engaging TO AT mode:

1. TO AT must be previously armed.
 2. TLA must be moved past the TO/GA position.
- TO AT engagement is confirmed by a green **TO** on the FMA.

AUTOMATIC AUTOTHROTTLE MODE TRANSITIONS:		
At 60 kts	TO automatically changes to HOLD mode, which holds max thrust.	
	Flight plan <u>not</u> loaded:	Flight plan <u>loaded</u> :
At 1200 ft	HOLD automatically switches to the SPD_T AT mode.	HOLD automatically switches to the SPD_E AT mode.

2. LATERAL AND VERTICAL GUIDANCE TAKEOFF MODES (TO)

- Represented on the **right** side of the FMA.
- Represented by crossbars FD and is used during the takeoff phase to maintain a pitch attitude.
- Deactivated when another vertical mode is selected.

Engaging TO mode:

- Engaged by pressing the TO/GA button on the throttle lever.
- **ROLL** and **TO** appears on the right side of the FMA. ROLL is the lateral mode, and TO is the vertical mode.
- If the flightplan has been loaded, white **LNAV** and **VNAV** modes will be indicated as armed.

AUTOMATIC LATERAL & VERTICAL MODE TRANSITIONS:		
At 100 kts	ROLL mode changes to TRACK lateral guidance mode which will hold the current track.	
	Flight plan <u>not</u> loaded:	Flight plan <u>loaded</u> :
At 400 ft	No action.	At 400ft, TRACK lateral mode automatically switches to LNAV mode.
At 1200 ft	TO automatically switches to FPA mode. TRACK remains the lateral mode until a different lateral mode has been selected.	TO vertical mode automatically switches to VNAV flight level change mode FLCH .

GO-AROUND MODE

- Represented on the **right** side of the FMA.
- Represented by crossbars FD and is used during the Go Around phase to maintain a pitch attitude.
- Engaged by pushing the TO/GA button during approach.
- Deactivated when another vertical mode is selected.

AUTOTHROTTLE

AT sets the correct engine thrust with the synchronized N1 for its various operating modes.

Normal disengagement:

- AT disconnect button on thrust lever or
- AT button on the guidance panel.

Automatically disengaged when one of the following occurs:

- After airplane touchdown
- Thrust levers beyond the TO/GA position (**Note:** this is implemented only with hardware throttles)

AUTOTHROTTLE MODES:

Takeoff mode - TO

- Armed on the ground by pressing the AT button on the guidance panel.
- Engages when thrust levers are at a high TLA.
- TLA NOT TOGA will be displayed on the CAS if the thrust levers are not in the TOGA detent by 60 kts.

Takeoff hold mode - HOLD

- Prevents undesired thrust lever movement during the takeoff phase.
- Engaged at or above 60 KIAS during the takeoff roll.

Speed on thrust mode SPD_T

- Controls the selected airspeed adjusting the engine thrust by moving the thrust levers.
- Vertical modes related to SPDT: FPA, VS, GS, PTH, GP, ALT, ASEL
- SPDT is the active mode when the FD is turned OFF (no active modes on FMA).

Speed on elevator - SPD_E

- AT maintains a fixed thrust setting, and the flight director adjusts the pitch attitude to maintain the selected airspeed.
- Vertical modes related to SPDE: FLCH, OVSP, VNAV flight level change (FLCH magenta)

Retard mode - RETD

- On the real plane, RETD reduces the thrust levers to idle during flare on landing when below 30 ft. However, we discovered that this mode was causing an issue for some users that have hardware throttles.

Therefore RETD currently works in the following way:

- During flare below 30 ft, RETD starts flashing, notifying the pilot that he needs to reduce the thrusts manually. AT gets disengaged

After discussing it with real pilots, this was the best compromise we could come up with, that would ensure the throttles get reduced for all users, with or without hardware throttles.

Go Around mode - GA

- Automatically advances the thrust levers to the TO/GA position.
- Engaged with the TO/GA button on the throttle levers.

Limit mode - LIM

- Autothrottle authority is not sufficient to achieve or maintain the selected airspeed.

Override mode - OVRD

- AT can be overridden by moving the thrust lever in any direction without causing its disengagement.
- **Note:** This only works if the user uses hardware throttles.

FMS SPEED

The FMS Speed is a function of the FMS that feeds specific speeds directly to the autothrottle. FMS Speed takes care of managing the speeds for you throughout the flight, from the takeoff to the landing, which is very convenient!

To engage FMS Speed, click on the switch located below the speed selector knob on the guidance panel. The options are **FMS Speed** and **Manual Speed**. This is only possible once you have entered the flight plan.



The FMS Speed must always follow the Speed Schedules (CLIMB/CRUISE/DESCENT) defined on PERF INIT 3/3 page, unless overwritten by any limits that would limit the speed to a lower value.

The list of limits is from the lowest to the highest priority. For example, the **Departure Speed Limit (4)** will override the **10,000ft Speed limit (3)** as well as **(2)** or **(1)**, but it cannot override **Flaps Speed limit (5)** or **Takeoff or Approach Speeds (6)** since those are higher priority.

1. Speed Schedule (Based on PERF INIT 3/3) - **Highest speeds**

Overriding Limits:

2. Manual waypoint speed entry (Defined by user)
3. 10000ft Speed Limit ("250 below 10000". Defined on the CLIMB Page)
4. Departure Speed Limit or GO Around Speed Limit (Limiting speeds around airports)
5. Flaps Speed limit (Set by the max flaps speed limits)
6. Takeoff or Approach Speeds - **Lowest speeds**

HOW FMS SPEED CHANGES THE SPEEDS

THESE CONDITIONS WILL TRIGGER...	THIS SPEED
TAKEOFF	
Takeoff Speed	V2 As entered by the pilot on the TAKEOFF 1/1 page on the FMS
Positive Rate of Climb	V2 + 10 KIAS
DEPARTURE	
- Above VNAV CAP AFE (VNAV Capture Above Field Elevation) 1,200 ft by default but can be changed on the DEPARTURE LIMIT FMS page	
- Above VNAV CAP AFE - Below the Departure AFE LIMIT	210 KIAS Speed is limited by the SPEED LIMIT defined on the DEP LIMIT Page
Above VNAV CAP AFE	Flap Speed Limit (VF) - 5 KIAS - VF is the maximum speed for the current flap configuration - VF changes as the pilot changes flap settings - FMS Speed will command → Flaps 1..... 225 KIAS → Flaps 2..... 210 KIAS → Flaps 3..... 195 KIAS → Flaps 4..... 175 KIAS → Flaps 5..... 175 KIAS → Flaps Full..... 160 KIAS
Speed restriction	Defined by procedure

• Flaps Clean • Above departure AFE LIMIT • Below 10,000ft	250 KIAS
CLIMB	
• Above 10,000 ft	Speed gradually increases to the CLIMB SPEED defined on PERF INIT 3/3
CRUISE	
• Top of Climb	CRUISE SPEED as defined on PERF INIT 3/3
DESCENT	
• One minute before TOD	DESCENT SPEED as defined on PERF INIT 3/3
• At 12,000ft	Gradually slows down to 250 KIAS prior to reaching 10,000ft.
APPROACH SPEEDS	
• 30NM out we enter the Terminal Area and Approach Speeds become available • Approach speeds don't have to be activated immediately as you would be flying too slow for too long. • Activate when ready, usually at 20NM out.	210 KIAS • If activated via the ACT APP SPEED prompt on the FMS 180 KIAS • If activated by deploying flaps the flaps
• Flaps 1 fully deployed	180 KIAS
• Flaps 2 fully deployed	160 KIAS
• Flaps 3 fully deployed	150 KIAS
• Flaps 4 fully deployed	140 KIAS
• Flaps 5 or FULL fully deployed	VAP
GO AROUND	

• When TOGA is pressed	VF - 5 KIAS • Same as TO after VNAV CAP • VF changes as the pilot changes flap settings • Speed is capped by the SPEED LIMIT found on the GO-AROUND LIMIT page
--	---

VF SPEEDS

On Departure:

On flaps 4, the VF bug will be 170
On flaps 3, the VF bug will be 190
On flaps 2, the VF bug will be 205
On flaps 1, the VF bug will be 220

On Arrival

On flaps clean, the VF bug will be 210
On flaps 1, the VF bug will be 180
On flaps 2, the VF bug will be 160
On flaps 3, the VF bug will be 150
On flaps 4, the VF bug will be 140
On flaps 5, the VF bug will be 140
On flaps full, the VF bug will disappear

FMS SPEED DETAILED DESCRIPTION

TAKEOFF SPEEDS

Takeoff Roll:

- When we select FMS Speed on the ground, the speed bug should change to magenta, and it will show the V2 value.
- V2 will be the bugged speed throughout the takeoff roll and after the takeoff
- PFD should show yellow dashes until V2 has been selected
- **You must enter the V2 value** (calculated on the EFB) otherwise the FMS will set an incorrect takeoff speed.

Positive Rate Of Climb:

- As soon as there is positive rate of climb, the speed will change to V2+10 kts

DEPARTURE SPEEDS

At VNAV CAP Altitude and below AFE LIMIT:

- At VNAV CAP AFE, bugged speed will change to **VF-5kts**
- The **pilot must manually retract the flaps**, to which FMS Speed will react.
- If they never retract the flaps, the bugged speed would stay VF-5 for the duration of the entire flight.

TERMS:

DEPARTURE LIMIT Speeds on departure will never exceed the *SPEED LIMIT* defined on the *DEPARTURE LIMIT* Page on the FMS. This is designed to maintain the aircraft below the maximum speed for class D airspace.

VNAV CAP AFE (VNAV Capture Above Field Elevation) is defined on the DEPARTURE LIMIT page.

AFE LIMIT (Above Field Elevation Limit) is also set on the DEPARTURE LIMIT page

Example: SPEED LIMIT is 200 kts

If the pilot sets flaps 2, the bugged speed would be 205 (VF for Flap 2 is 210, minus 5 knots is 205). However, if they are still below AFE, the speed would be capped at the SPEED LIMIT of 200kts until we get above AFE.

ABOVE AFE LIMIT & BELOW 10000 FT

- Aircraft will maintain 250kts

CLIMB SPEEDS

- Once the aircraft is above 10000ft, FMS Speed will slowly increase speed up to the CLIMB SPEED defined on PERFORMANCE INIT 3/3
- The target is the lower value of the CAS/MACH climb speed schedule
- Changes to MACH are automatic only when using FMS Speed

CRUISE SPEEDS

- The speed command transitions to CRUISE SPEED as defined on PERFORMANCE INIT 3/3 when the aircraft is in **ALT**, or **ALT** and the aircraft altitude equals the PERFORMANCE INIT cruise altitude.
- When the aircraft levels off at an altitude below the CRUISE ALTITUDE, the speed command continues to be the climb speed command.

DESCENT SPEEDS

- When the aircraft begins descending below the cruise altitude prior to reaching 50 NM of the TOD, the speed command remains in cruise speed schedule.
- When the aircraft begins descending below the cruise altitude and is within 50 NM of the TOD, the speed command changes to the descent speed schedule and the descent CAS/MACH target is displayed.
- The descent speed command is used during any intermediate level offs.
- When the aircraft nears the speed/altitude limit (10000ft), the speed command anticipates the speed limit and slows the aircraft **prior to reaching the altitude.**

APPROACH SPEEDS

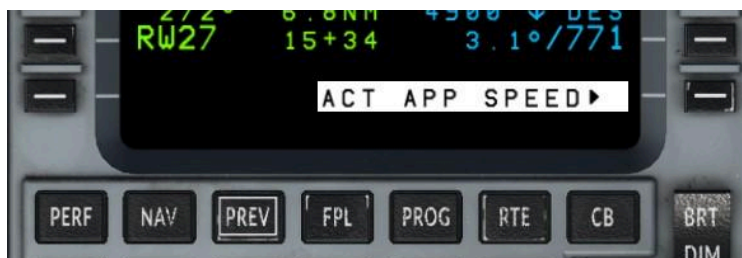
- The Approach speed schedule is defined on the APPROACH SPEEDS page. These speeds can be changed by the pilot.
- The speed will change based on the pilot manually deploying flaps.



ACTIVATING APPROACH SPEEDS

Activating Approach Speeds is done manually by either selecting the active approach speeds prompt on the FMS or by extending flaps

- 30NM from the destination airport, we enter the “Terminal Area”, also announced by the magenta TERM annunciator on the PFD
- At that moment a reverse video **ACTIVATE APP SPD** is displayed on 6R on the FPL Page.
- Pressing it activates Clean Flaps speeds 210 kts.
- Another way to activate approach speeds is **deploying flaps**, when the speed will go directly to 180kts.



GO AROUND SPEEDS

- This mirrors the Departure Speeds takeoff phase except it uses the GO-AROUND LIMIT page for the SPEED LIMIT or AFE LIMIT.

WAYPOINT SPEED CONSTRAINT

- The FMS gives the ability to cross a waypoint at a specified speed. This is referred to as a waypoint speed constraint.
- A waypoint speed constraint is retrieved from a procedure (SID/STAR/approach) automatically or is entered by the pilot on the FLT page.

VERTICAL & LATERAL FLIGHT GUIDANCE

FLIGHT GUIDANCE MODE COLOR DESIGNATION

The color of the mode annunciator will define whether a mode is managed by the pilot or by the FMS.

- All **GREEN** modes are manual/managed
- All **MAGENTA** modes are controlled by the FMS

LATERAL MODES

Roll **ROLL**

- Holds the selected bank (roll) angle
- Roll hold is the basic lateral mode and it is activated when:
 - The active lateral mode is deactivated.
 - The TO mode is selected on ground by pressing TOGA button.
 - The A/P is engaged with no other FGCS modes active.

Heading **HDG**

- The heading select mode activates when the HDG pushbutton on the GP is pressed.
- The heading select mode is deactivated when:
 - HDG pushbutton is pressed a second time.
 - A different lateral mode is selected.
 - Armed LNAV, LOC or BC modes are activated.
- Pressing the HDG selector knob synchronizes the heading bug to the current heading. With HDG mode active, the FD follows the selected heading and respects the side to which the turn was commanded if the turn is greater than 180 degrees.

Lateral Navigation **LNAV**

- The FD provides lateral guidance commands for interception, capture, and tracking of FMS-generated courses.

- The LNAV mode is selected when the NAV button is pressed on the guidance panel with the FMS as the source of navigation displayed on the PFD.
- The FD will not arm or track a VOR signal in LNAV. VOR tracking must be done manually in HDG mode.

Localizer **LOC**

- Localizer mode guidance is computed based on PFD data. The LOC mode is selected via the NAV or APP buttons on the GP.
- The APP button will arm the LOC (and GS) modes if a valid localizer frequency is tuned and either PREV or V/L is selected on the PFD.
- The NAV button is selected when the GS mode is not possible or not desired.

Track Hold **TRACK**

- The track hold mode maintains an inertial-derived aircraft track from the IRS.
- This mode is engaged automatically when GA or TO is selected by the TOGA.
- The automatic transition from ROLL to TRACK occurs when:
 - IAS is greater than 100 kt for TO or greater than 100 kt for GA.
 - And bank angle is at 3° or below for more than 10 seconds.
- Selecting another lateral flight director mode disengages track mode.

VERTICAL MODES

Flight Path Angle **FPA**

- Flight path angle ($\pm 9.9^\circ$) is selected by the FPA SEL selector knob on the guidance panel. This mode will hold the selected pitch.
- The FPA is the basic vertical mode. FPA mode becomes the active FD mode when:
 - The FPA pushbutton on the guidance panel is pressed.
 - Engaging the autopilot if no FD mode is active.
 - When a lateral mode is activated and there is no vertical flight director guidance mode active.
 - Deselecting the current vertical mode.

Takeoff **TO**

- The takeoff mode is a FD-only mode. In TO mode, a dual-cue flight director (crossbar) is presented on the PFD.
- Lateral mode changes are allowed while the active vertical mode is TO.
- The Flight Path Vector (FPV) is not displayed while the TO mode is active.
- After takeoff, the Flight Director increases pitch automatically based on the commanded speed as soon as the aircraft leaves the ground.

Flight Level Change **FLCH**

- The FLCH mode provides pitch guidance to climb or descend at the speed which has been selected via the Speed Selector Knob.
- FLCH guidance is associated with the SPDE autothrottle mode.
- The selected speed is displayed in the box on top of the speed tape.
- The FLCH button on the GP activates the FLCH mode. The FLCH mode deactivates when another vertical mode is activated.
- Although FLCH mode attempts to hold the selected speed with pitch, it will not cause a vertical reversal; i.e., it will not command a climb when a lower altitude is selected, and it will not command a descent when a higher altitude is selected.

Altitude Hold **ALT**

- The altitude hold mode maintains the altitude existing upon activation.
- ALT mode is automatically activated after ASEL successfully captures a selected altitude.
- After ALT mode engagement at the selected altitude, change to another vertical mode is possible only after selecting a different altitude via the ALT SEL knob, or by de-selecting ALT with the ALT button on the Glareshield Panel(GP). This logic is valid for all vertical modes, except for Glide Slope (GS) and Go-Around (GA). Switching from ALT mode to GS mode may occur without change in ALT SEL selection. Also, switching from ALT mode to GA mode will occur upon activation of the TOGA button without change of ALT SEL selection.
- Altitude Hold can also be activated manually by pressing the ALT button on the GP. In this case, ALT holds the altitude existing at the moment the button was pressed, not the altitude set in the pre-select window.

Vertical Speed **VS**

- The VS mode maintains a vertical speed.
- The VS mode is activated by pressing the VS button on the GP.
- Vertical speed is selected by rotating the vertical speed thumbwheel on the GP.
- The vertical speed command range goes from -8000 ft/min. to +6000 ft/min.

Overspeed **OVSP**

- Flight Director provides overspeed protection during FLCH, VS and FPA modes.
- The OVSP protection limits the aircraft speed to VMO/MMO by commanding pitch adjustments to limit the airspeed.
- When the OVSP protection activates, an amber OVSP indication displays on the FMA. The previous active mode is displayed as armed (white) and becomes active again when OVSP protection is no longer active.

Glideslope **GS**

- The GS mode arms when the APP button is pressed and activates when the glide slope is captured.
- GS mode can only be active when LOC is the active lateral mode (i.e., the localizer must be captured before the glideslope can be captured).
- GS will deactivate if a different lateral mode is selected.

Go Around **GA**

- GA mode will maintain an 8° pitch attitude

VNAV Path **PTH**

- The VNAV Path mode provides vertical guidance to intercept and track an FMS-generated vertical path descent to meet a waypoint altitude constraint.
- For more information, see the FMS manual and the VNAV section.

Vertical Navigation VNAV

- VNAV can be engaged manually by pressing VNAV button on the GP
- VNAV engages automatically at 1200ft if the flight plan has been loaded and TOGA was pressed before takeoff (arming LNAV&VNAV).
- For more information, see the FMS manual and the VNAV section.

How to use VNAV

[Please visit our website](https://www.xcrafts.com/vnav) for a detailed description of different VNAV modes and when and how you should use them: <https://www.xcrafts.com/vnav>

AIRPLANE GENERAL

COCKPIT CONCEPT

The cockpit uses Dark & Quiet philosophy, therefore if everything is running as it should, all the buttons will be dark and quiet.

When a white stripe is illuminated in a button, it indicates the button is not in its normal position.



OVERHEAD PANEL



DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL		APU CONTROL PANEL		
	WIPERS PANEL	AIR COND / PNEUMATIC PANEL		
	FUEL PANEL		EXTERNAL LIGHTS	PRESSURIZATION PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	PAX OXYGEN PANEL		

DVDR CONTROL PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
		WIPERS PANEL		
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL

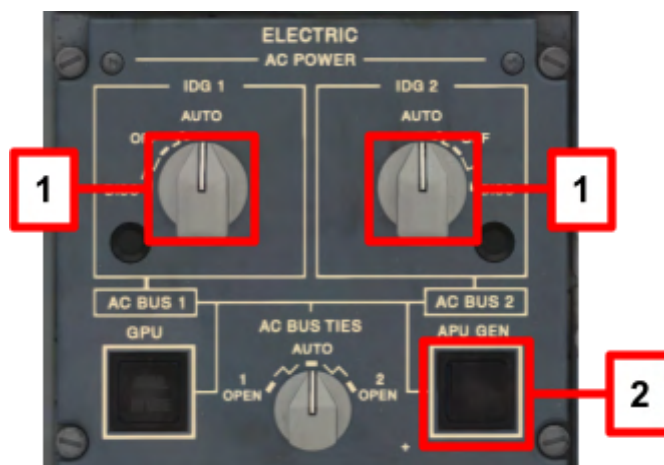
- The digital voice data recorder system (DVDR) combines a flight data recorder (FDR) and a cockpit voice recorder (CVR) in a single unit.
- **Note:** This panel is not simulated in the current version.



ELECTRICAL PANEL

AC POWER

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
AC POWER		APU CONTROL PANEL		
DC POWER	FUEL PANEL	WIPERS PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL

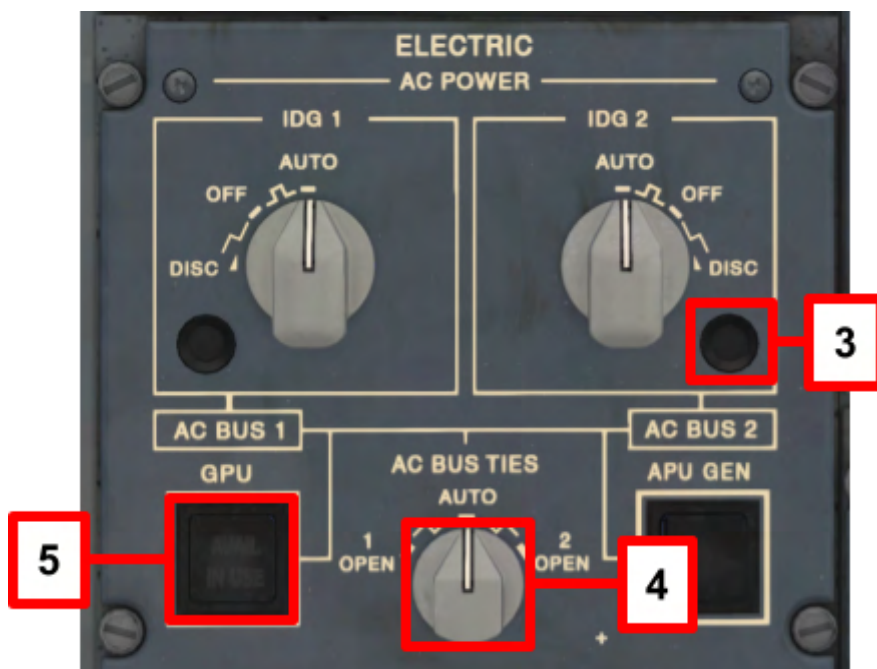


1. IDG (INTEGRATED DRIVE GENERATOR) 1 and IDG 2 SELECTOR KNOBS

- One IDG can supply the entire electrical system
- **AUTO**: Closes the IDG contactor, connecting the IDG to the respective AC BUS. When an engine starts, the IDG will automatically come on line, powering the respective AC BUS. The previous power source is disconnected from that AC BUS.
- **OFF**: Opens the IDG contactor, isolating the IDG from the respective AC BUS.
- **DISC (Not simulated in this version)** This is used on the real aircraft to mechanically disconnect the IDG.
 - Source priority for the AC BUS 1 and 2: Respective IDG (Onside), APU Generator (Inside), Ground Power Unit (Outside), Opposite IDG (Cross-side)

2. APU GENERATOR

- APU generator has the same power rating as the IDGs.
- **PUSH IN (DARK)**: Automatically connects the APU Generator to the AC BUS TIE, according to the source priority.
- **PUSH OUT (WHITE STRIPE)**: Opens the APU Generator contactor and isolates the APU generator from the AC BUS TIE.



3. IDG OIL INDICATOR

- Illuminates for low/high oil pressure, indicating which IDG must be disconnected (**Note:** This feature is not simulated in the current version)
- Illuminates during ANNUNCIATOR TEST

4. AC BUS TIE SELECTOR KNOB

- **1 OPEN:** Opens the BUS Tie Contactor (BTC) 1, segregating AC BUS 1 from AC BUS 2.
- **AUTO:** This position controls the operation of the BTCs, according to system logic. If one of the AC power sources fails, both AC electrical networks are automatically connected by BTCs.
- **2 OPEN:** Opens the BUS Tie Contactor (BTC) 2, segregating AC BUS 2 from AC BUS 1.

5. GROUND POWER UNIT (GPU BUTTON)

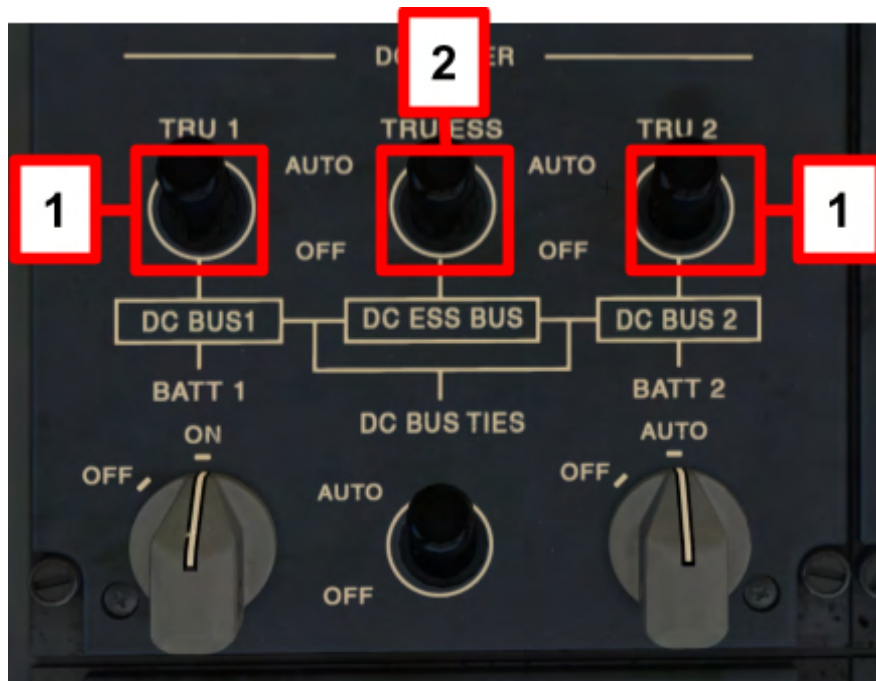
- **DARK:** The GPU is not connected
- **AVAIL:** The GPU is connected to the airplane, but not to the AC BUS TIE
- **IN USE:** AC GPU is powering the airplane.
- If the APU Generator is providing power to the AC BUS TIE, GPU will disconnect from the AC BUS TIE and the button will go from IN USE to AVAIL according to system logic

GROUND POWER UNIT (GPU)

- The GPU can be connected through the EFB Ground Services page.
- If the GPU is connected and the GPU button shows IN USE, the GPU provides AC power to the electrical busses.

DC POWER

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
AC POWER	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
DC POWER		WIPERS PANEL		
COCKPIT LIGHTS		EXTERNAL LIGHTS		
	PAX SIGNS PANEL		PRESSURIZATION PANEL	PAX OXYGEN PANEL

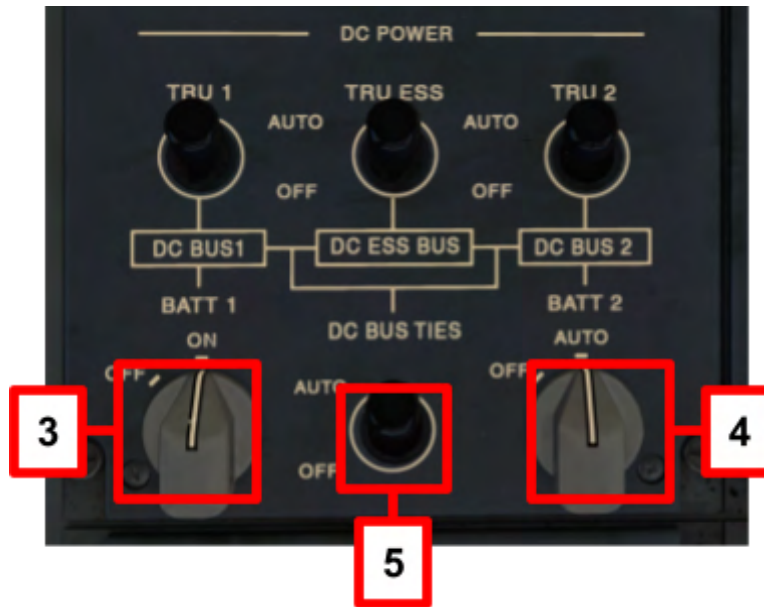


1. TRU (TRANSFORMER RECTIFIER UNIT) 1 AND TRU 2 TOGGLE SWITCH

- Converts AC power to DC power
- AUTO: Automatic operation of the electrical system. Connects TRU 1 (2) to the respective DC BUS
- OFF: Isolates TRU 1 (2) from the respective DC BUS

2. TRU ESS SWITCH

- AUTO: Automatic operation of the electrical system. Connects TRU ESS to the ESS BUS 3
- OFF: TRU ESS is isolated from ESS BUS 3



3. BATT 1 SELECTOR KNOB

- ON: Connects Battery 1 to DC ESS BUS 1
- OFF: Disconnects Battery 1

4. BATT 2 SELECTOR KNOB

- ON: Connects Battery 2 to DC ESS BUS 2
- OFF: Disconnects Battery 2

5. DC BUS TIES

- AUTO: Allows automatic operation of the electrical system.
- OFF: DC BUS TIES disconnected

ELECTRICAL SYSTEM

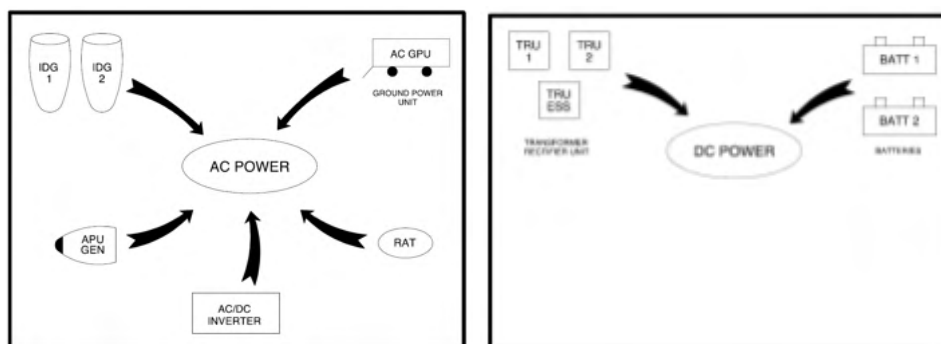
- Two independent networks provide complete segregation and fault isolation
- GPU-powered aircraft with an engine started at the gate will have a split electrical system

AC ELECTRICAL SYSTEM

- **Two IDGs, APU generator, RAT, AC/DC inverter, AC GPU**
- Once AC power is established, all displays, panel lights, cockpit lights and instruments are operational

DC ELECTRICAL SYSTEM

- Three TRUs, Two batteries,
- DC GPU is not simulated.



INITIAL POWER-UP / ELECTRICAL EMERGENCY

If the airplane is only powered from the DC Electrical system, for example during power up or in an electrical emergency, the following items are operational:

- Guidance panel, MFD (CA side), EICAS, IESS, Clock, MCDU 2, CCD1, Audio control panels and compass backlight.

COCKPIT LIGHTS PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. MAIN PNL KNOB

- OFF: Turns the lights off.
- BRT: lights full bright.
- Rotate knob to adjust brightness of the main panel integral lighting.

2. OVHD PNL KNOB

- OFF: Turns the lights off.
- BRT: lights full bright.
- Rotate knob to adjust brightness of the overhead panel integral lighting.

3. PEDESTAL KNOB

- OFF: Turns the lights off.
- BRT: lights full bright.

- Rotate knob to adjust brightness of the pedestal panel integral lighting.

4. ANNUNCIATORS TEST BUTTON (Momentary position)

- Illuminates the white/amber striped bars and caption indications verifying the lamp integrity.
- Does NOT check Parking brake ON light, AC GPU Button, or Fire lights

5. DOME SWITCH

- ON: Turns 2 cockpit dome lights ON
- OFF: Turns 2 cockpit dome lights OFF

ENGINE FIRE HANDLE

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



ENGINE FIRE HANDLE

- PULL 1: Discharges the A (left) fire extinguisher bottle into the associated engine
- PULL 2: Discharges the B (right) fire extinguisher bottle into the associated engine

Note: On the real aircraft, both handles can extinguish either fire bottle by rotating in the associated direction. This is not simulated in this version.

ENGINE FIRE INDICATION

- Engine Fire handle illuminates
- EICAS displays ENG (1) (2) FIRE message
- Master Warning triple chime and Fire bell
- **FIRE** displayed on the respective engine ITT gauge
- Master Warning lights on glareshield illuminate

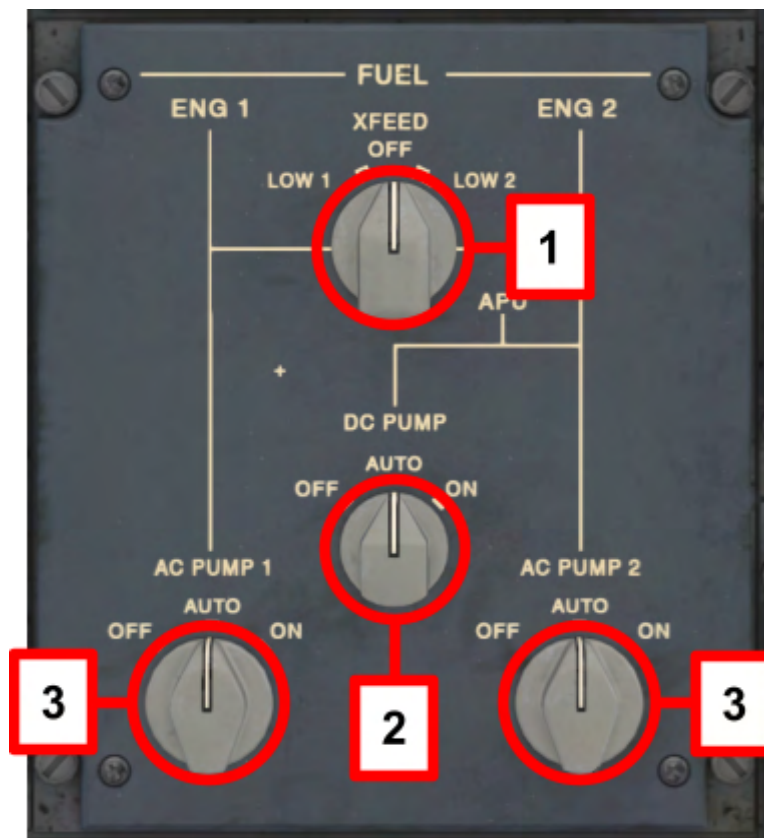


FUEL PANEL

Fuel System Overview

The fuel system manages fuel distribution between tanks and engines, allowing for crossfeed, auxiliary tank isolation, and fuel pump operation. Below are the key switches and their functions:

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. XFEED SELECTOR KNOB

- LOW 1: Opens the crossfeed valve and automatically activates AC PUMP 2, supplying fuel to both engines from the right wing tank
- OFF: Closes the crossfeed valve
- LOW 2: Opens the crossfeed valve and automatically activates AC PUMP 1, supplying fuel to both engines from the left wing tank

2. DC PUMP SELECTOR KNOB

- OFF: Deactivates the DC Fuel pump
- AUTO: Automatically operates the DC fuel pump according to system logic
- ON: Activates the DC fuel pump

3. AC PUMP SELECTOR KNOBS

- OFF: Deactivates the associated AC Fuel pump
- AUTO: Automatically operates the associated AC fuel pump according to system logic
- ON: Activates the associated AC fuel pump

Lineage 1000 Specific features

The fuel system is designed to continuously provide fuel to the engines and APU. Fuel is stored in two integral wing tanks and in two auxiliary tanks. The auxiliary tanks are installed in the pressurized forward and aft baggage compartments in the fuselage and provide fuel to the wing tanks.

Crossfeed System (Xfeed)

- **Xfeed Override Button**
 - Activates the override function (only works if the cap is open).
- **Xfeed Switch:** Controls fuel crossfeed between tanks.
 - **Positions:** OFF, AUTO, ON

Auxiliary Fuel System

- **AUX Tanks Isolation Valves Button:** Opens/closes AUX tank valves (only works if the cap is open).
- **AUX Fuel Transfer Switch:** Controls the transfer of fuel from auxiliary tanks.
 - **Positions:** OFF, AUTO, ON

PASSENGER SIGNS PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. EMER LT SELECTOR KNOB

- OFF: Prevents all emergency lights from illuminating if electrical power is turned off or fails.
- ARMED: Automatically illuminates all emergency lights if the airplane loses electrical power or if airplane electrical power is turned off
- ON: Turns on all emergency lights

2. ATTND CALL

- PUSH: Single Hi/Lo tone chime in passenger cabin

3. STERILE SWITCH

- ON: Illuminates the sterile rainbow light in the cabin
- OFF: Turns off the sterile rainbow light in the cabin

4. NO ELEC DEVICES / NO SMKG SWITCH

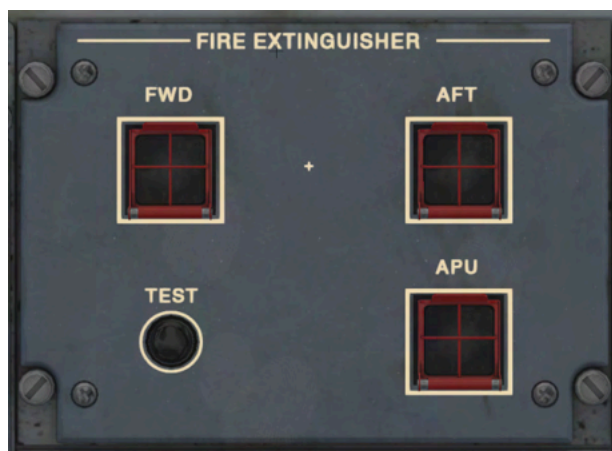
- ON: Illuminates the respective signs
- OFF: Turns off the respective signs

5. FSTN BELTS SWITCH

- ON: Illuminates the FASTEN SEAT BELTS signs
- OFF: Turns off the FASTEN SEAT BELTS signs
- This switch will control automatic stowing of table trays, window shades or cabin doors on the Lineage 100

FIRE EXTINGUISHER PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
		WIPERS PANEL		
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. FWD/AFT CARGO FIRE DETECTION/EXTINGUISHER BUTTON (Guarded)

Button illuminates if smoke is detected in the respective cargo compartment.

Note: Cargo compartment fire is not simulated in this version.

2. FIRE DETECTION SYSTEM TEST BUTTON (Momentary position)

- Tests the engine fire, APU fire, and cargo smoke detection systems
- Press and hold the button for a minimum of 3 seconds
- **Note:** Automatic APU Shutdown after holding for 10 seconds is currently not simulated

FIRE DETECTION SYSTEM TEST

- Displays fire detection/extinguishing system messages on EICAS

The following warnings happen during a FIRE DETECTION SYSTEM TEST:

OVERHEAD PANEL (6 lights)

- Fire Handle 1
- Fire Handle 2
- Cargo smoke (FWD) extinguisher button
- Cargo smoke (AFT) extinguisher button
- APU Extinguisher button
- APU EMER STOP (top half of button)

GLARESHIELD (2 lights)

- Master Warning Light (Left)
- Master Warning Light (Right)

EICAS (7 lights)

- ENG 1 FIRE (CAS message)
- ENG 2 FIRE (CAS message)
- CARGO AFT SMOKE (CAS message)
- CARGO FWD SMOKE (CAS message)
- APU FIRE (CAS message)
- **FIRE** displayed on the respective engine ITT gauge

AURAL WARNING (2 aural warnings)

- Master Warning Triple Chime
- Fire Bell

3. APU FIRE EXTINGUISHER BUTTON (Guarded)

- **Note:** APU fire is not simulated in this version of the product

APU CONTROL PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. APU SELECTOR KNOB

- OFF: Normal shutdown is initiated. Electric power is removed at the end of a 1-minute cooldown. Rotating the knob back to ON during shutdown cancels the shutdown sequence
- ON: Normal position when running
- START (Momentary position): Initiates the APU start cycle. The knob can be turned to START position only after the Yellow box on the APU section of EICAS stops flashing (Around 3 seconds).

2. APU EMERGENCY STOP BUTTON

- **Note:** This function is not simulated in this version of the product. On the real aircraft, this button shuts down the APU without cooldown

AUXILIARY POWER UNIT (APU)

- Gas turbine engine that provides AC power and bleed air to the airplane

APU START

- To avoid unsuccessful start attempt it is recommended to wait 30 seconds after the EICAS is energized to start the APU.
- APU electrical and pneumatic power are available after reaching ~95% rpm.
- AC GPU cannot be used to start the APU

APU BLEED

- Primarily used for: ECS on the ground, engine starting on the ground, assist with engine start ($\leq$ FL210), and bleed source for pressurization, and heating and cooling ($\leq$ 15000 MSL)

WINDSHIELD WIPER PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. WINDSHIELD WIPER

- TIMER: Not simulated in this version of the product.
- OFF: Windshield wiper stops in the stowed position
- LOW: Operates at a low speed
- HI: Operates at a high speed

EXTERNAL LIGHTS PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
		WIPERS PANEL		
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. NAV SWITCH

- ON: Nav lights on
- OFF: Nav lights off

2. STROBE SWITCH

- ON: Strobe lights on
- OFF: Strobe lights off

3. RED BCN SWITCH

- ON: Red beacon lights on
- OFF: Red beacon lights off

4. LOGO SWITCH

- ON: Logo lights on
- OFF: Logo lights off
- Logo lights are located on the horizontal stabilizer and illuminate the logo on the vertical stabilizer

5. TAXI LIGHT SWITCHES

- ON: Respective taxi light on
- OFF: Respective taxi light off

6. INSP LIGHT SWITCHES

- ON: Wing inspection lights on
- OFF: Wing inspection lights off
- Inspection lights are mounted on either side of the fuselage forward of the wing. They illuminate the engine intakes and the wing leading edges.

7. LANDING LIGHT SWITCHES

- ON: Respective landing light on
- OFF: Respective landing light off

HYDRAULIC PANEL

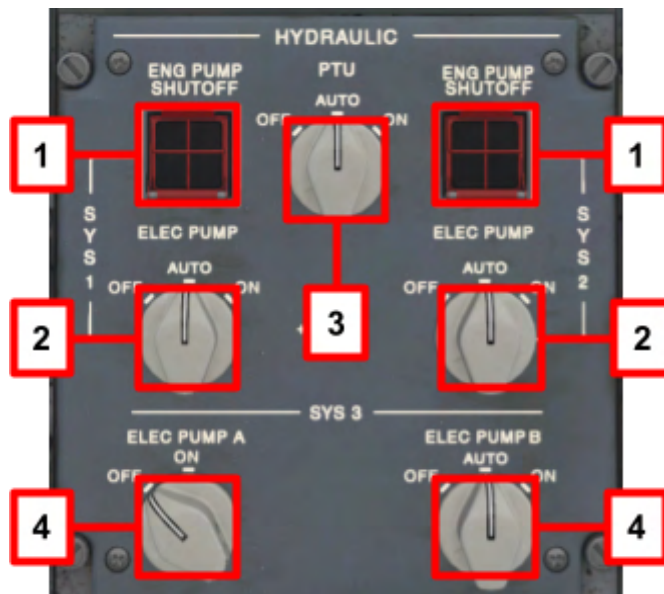
DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
		WIPERS PANEL		
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL

HYDRAULIC SYSTEM

- 3 independent systems
- **Note:** System 3 is not simulated properly due to limitations of X-Plane. Future updates of X-Plane will hopefully allow us to also simulate a third hydraulic system properly.
- Primary and secondary flight controls, landing gear, brakes, nosewheel steering, thrust reversers and spoilers.

HYD SYS 1 & 2

- Normal operating pressure 3000 psi
- Engine Driven Pump (EDP) is primary and the AC Electric hydraulic pump is backup
- Engine Driven Pump (EDP) cannot be disabled



1. ENG PUMP SHUTOFF BUTTON (Guarded)

- **Note:** This function is not simulated in this version of the product

2. ELECTRIC HYDRAULIC PUMP SELECTOR KNOBS

- OFF: Turns the respective pump off
- AUTO: In this version of the product, AUTO acts as the ON position. Pump operates continuously.

3. PTU (POWER TRANSFER UNIT) KNOB

- **Note:** This function is not simulated in this version of the product

4. HYDRAULIC SYSTEM 3 SELECTOR KNOBS

- ON: Turns the respective pump on
- OFF: Turns the respective pump off
- AUTO (3B): Turns 3B on automatically if AC pump 3A fails. Note that this is not simulated in this version of the product.

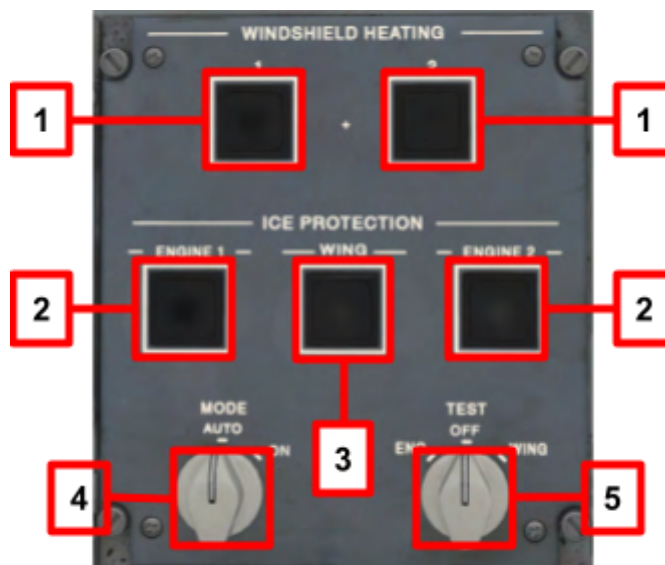
PRESSURIZATION PANEL

- **MODE Switch**
 - **MAN:** Enables the use of the CABIN ALT switch to manually adjust the cabin altitude.
 - **AUTO:** Automatic operation
 - **LFE CTRL:** This features is not currently implemented. It allows for manual adjustment of the Landing Field Elevation
- **CABIN ALT switch:** Use this switch to manually adjust the cabin altitude if the MODE is MAN. Please note that the switch is not animated yet, however it is operational.



WINDSHIELD HEATING / ICE PROTECTION PANEL

DVDR CONTROL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXTINGUISHER PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CONTROL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
		WIPERS PANEL		
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. WINDSHIELD HEATING (1) (2) BUTTON:

- PUSH IN (Dark): Activates the respective heating system
- PUSH OUT (White stripe): Deactivates the respective heating system
- Buttons are not independent, they both operate windshield heating of the entire windshield in this version of the product.

2. ICE PROTECTION ENGINE (1)(2) BUTTON:

- PUSH IN (Dark): Activates the respective engine anti-ice system
- PUSH OUT (White stripe): Deactivates the respective engine anti-ice system

3. WING ICE PROTECTION BUTTON

- PUSH IN (Dark): Activates the respective heating system
- PUSH OUT (White stripe): Deactivates the respective heating system.

4. ICE PROTECTION MODE SELECTOR KNOB

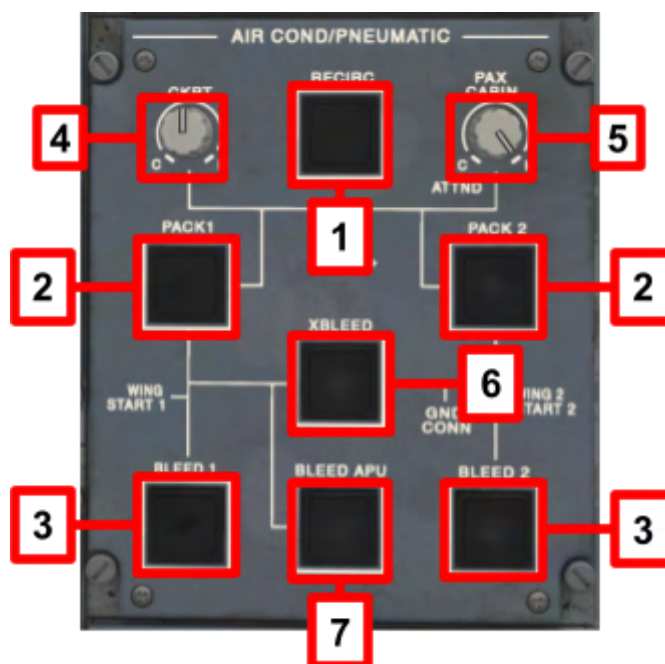
- AUTO: In this version of the product, AUTO acts as the ON position
- It is not possible to rotate the switch to the ON position as it has the same function in the AUTO position.

5. ICE PROTECTION TEST SELECTOR KNOB

- **Note:** This function is not simulated in this version of the product

AIR COND / PNEUMATIC PANEL

DVDR CTRL PANEL	ENGINE 1 FIRE HANDLE	FIRE EXT PANEL	ENGINE 2 FIRE HANDLE	WINDSHIELD HEATING / ICE PROTECTION
ELECTRICAL PANEL	FUEL PANEL	APU CTRL PANEL	HYDRAULIC PANEL	AIR COND / PNEUMATIC PANEL
COCKPIT LIGHTS	PAX SIGNS PANEL	EXTERNAL LIGHTS	PRESSURIZATION PANEL	PAX OXYGEN PANEL



1. RECIRC BUTTON

- PUSH IN (Dark): Turns on both recirculation fans, according to system logic
- PUSH OUT (White stripe): Turns off both recirculation fans

2. PACK (1) (2) BUTTON

- PUSH IN (Dark): Opens respective pack flow control valve
- PUSH OUT (White stripe): Manually closes the respective pack flow control valve

3. BLEED (1) (2) BUTTON

- PUSH IN (Dark): Commands the respective engine bleed valves, according to system logic
- PUSH OUT (White stripe): Manually closes the respective engine bleed valves

4. CKPT TEMPERATURE SELECTOR

- Controls cockpit temperature relative to the knob position

5. PAX CABIN TEMPERATURE SELECTOR

- Controls cabin temperature relative to the knob position

6. XBLEED BUTTON

- PUSH IN (Dark): Opens the cross-bleed valve, according to system logic
- PUSH OUT (White stripe): Manually closes the cross-bleed valve

7. BLEED APU BUTTON

- PUSH IN (Dark): Commands automatic operation of the APU bleed valve, according to system logic.
- PUSH OUT (White stripe): Manually closes the the APU bleed valve

PASSENGER OXYGEN PANEL

- **Note:** This function is not simulated in this version of the product



GLARESHIELD PANEL



GLARESHIELD LIGHTS / COMMUNICATION CONTROL PANEL



1. COMP/CLK LIGHT KNOB

- Regulates the brightness of the compass/clock lighting

2. MFD, PFD and EICAS LIGHT KNOB

- Regulates the brightness of the MFD, PFD or EICAS

3. FLOOD/STORM LIGHT KNOB

- Turns on/off and regulates the brightness of the flood/storm panel lighting

4. CHART LIGHT KNOB

- Turns on/off and regulates the brightness of associated chart holder lighting

5. PTT BUTTON

- **Note:** This function is not simulated in this version of the product

MASTER CAUTION/WARNING PUSHBUTTONS



1. MASTER WARNING PUSHBUTTON

- PUSH: Acknowledges the warning messages and extinguishes the light

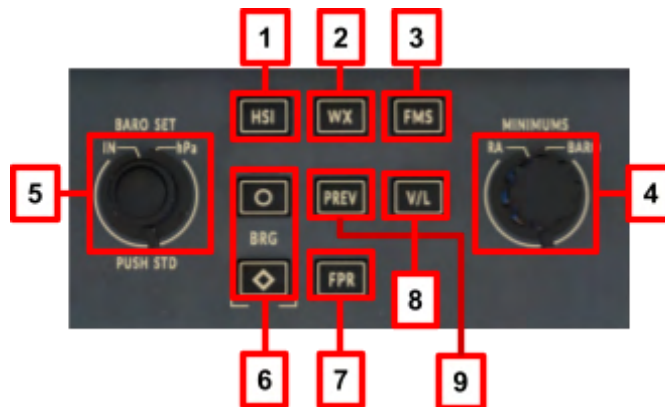


2. MASTER CAUTION PUSHBUTTON

- PUSH: Acknowledges the caution messages and extinguishes the light



DISPLAY CONTROLLER PANEL (LEFT/RIGHT)



1. HORIZONTAL SITUATION INDICATOR (HSI) BUTTON

- Selects between full compass, arc, or map mode for the PFD

2. WEATHER RADAR (WX) BUTTON

- Displays WX Radar information on PFD (HSI in map mode only)

Below you can see all the possible HSI modes on the PFD:



3. FMS SELECTION / FMS SOURCES

- Selects FMS as the primary navigation source

4. MINIMUMS SELECTOR KNOB

- Outer Knob
 - RA: Sets radio altimeter minimums. Used for ILS CAT II Approaches
 - BARO: Sets barometric minimums. Used for ILS CAT I Approaches
- Inner Knob
 - Selects decision height (DH), decision altitude (DA) or minimum descent altitude (DA) based on position of RA/BARO selector knob

5. BAROMETRIC SET KNOB

- Inner Knob
 - Sets barometric altimeter correction
 - Pushing this control knob sets barometric correction to standard
- Outer Knob
 - IN: Sets barometric correction format to inches of mercury
 - hPa: Sets barometric correction format to HectoPascals

6. BEARING SOURCES (BRG) BUTTON

- CIRCLE (○): Allows VOR1, ADF1 selection for the PFD display
- DIAMOND (◆): Allows VOR2, ADF2 selection for the PFD display

7. FLIGHT PATH REFERENCE (FPR) BUTTON

- Commands the flight path reference line and the flight path digital readout
- **Note:** This function is not simulated in this version of the product

8. VOR/LOC BUTTON (V/L)

- Selects VOR or LOC as the primary navigation source for the PFD
- Toggles between VOR1/LOC1 and VOR2/LOC2 (NAV1/NAV2)

9. PREVIEW BUTTON

- Selects the preview mode when the FMS is the primary navigation source. The course and the lateral/vertical deviation can be previewed
 - Pressing the first time: enables the VOR1/LOC1 preview
 - Pressing the second time: enables the VOR2/LOC2 preview
 - Pressing the third time: disables the preview mode
- Search "ILS AUTOTUNE" to learn how the PREVIEW mode engages automatically.

GUIDANCE PANEL

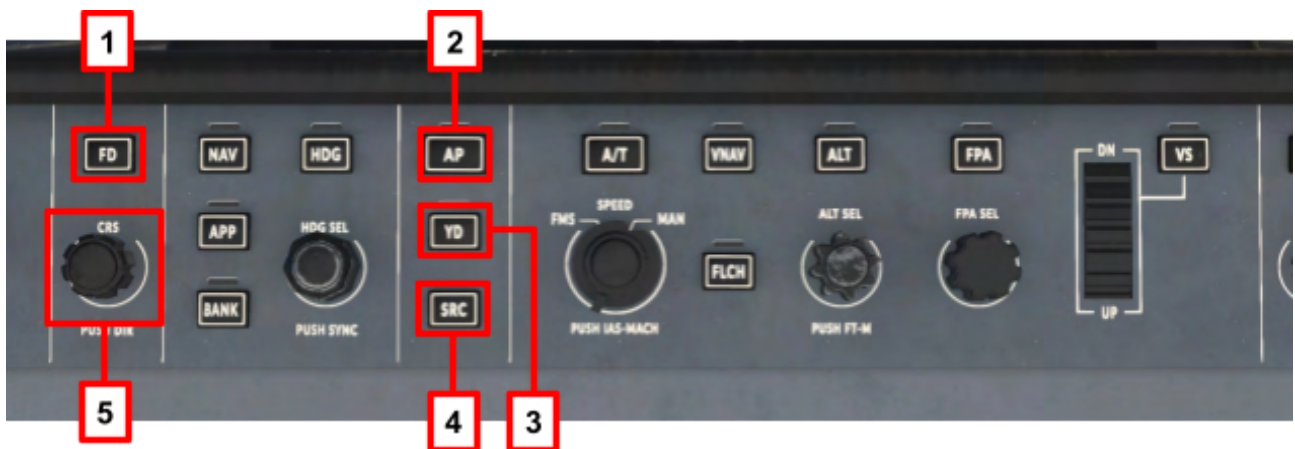
STATUS INDICATORS



Although, there are light indicators above the buttons on the GP, they **do not illuminate** to indicate armed or active status.

This is not a bug. Instead, it is an airline option to disable them and that is how it is on the majority of the real E-Jets.

The reason behind this is that the pilots should only ever look at the FMA to verify the AP status and keep their eyes on the PFD.



1. FD BUTTON

- Turns ON and OFF the Flight Director presentation on PFD
- If no modes are selected, the FD displays once a lateral or vertical mode is selected, or once TOGA is pressed on the throttles.

2. AP BUTTON

- Engages or disengages the Autopilot

3. YD BUTTON

- Engages or disengages the Yaw Damper

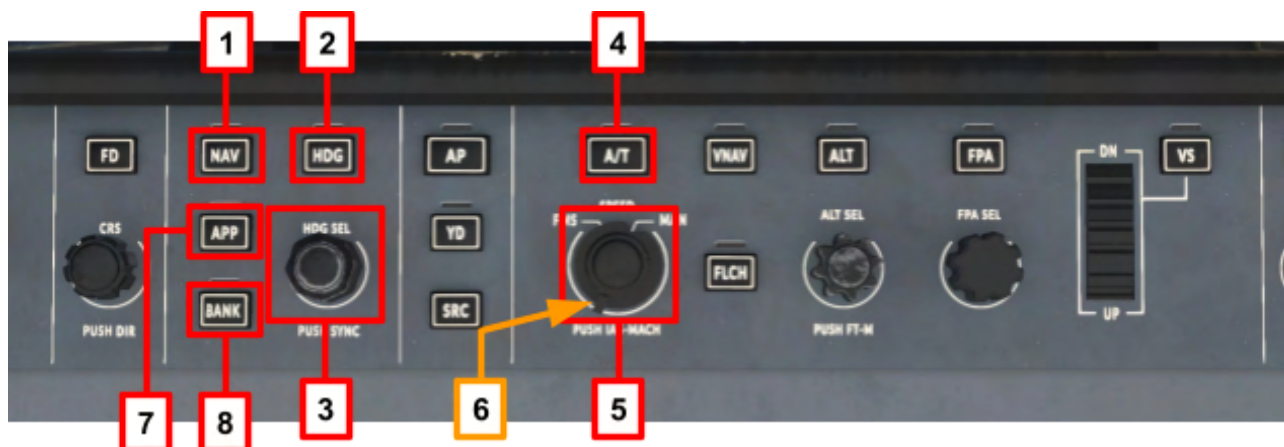
4. SRC BUTTON

- Alternates between the Captain and First Officer FMA modes as source for the autopilot. A green arrowhead on the FMA indicates the respective source selected.
- Note:** This function is not simulated in this version of the product

5. CRS BUTTON

- Manually sets the desired course on CDI as described below.
- CDI displaying V/L: Sets CDI course

SPEED MODECONTROLS AND LATERAL GUIDANCE CONTROLS



1. NAV BUTTON

- On the ground:
 - Button arms/disarms LNAV
- In the air:
 - According to the presentation selected for the CDI the NAV Button engages:
 - LOC: If the CDI is selected V/L (Green) and the source ILS frequency is tuned.
 - LNAV: If the CDI is selected FMS (Magenta)
 - Pressing NAV with LOC or LNAV engaged reverts the lateral guidance to ROLL

2. HDG BUTTON

- Selects and deselects the heading select mode

3. HDG SELECTOR KNOB

- Manually selects the desired heading
- Pressing this knob synchronizes the heading select to the current heading

4. A/T BUTTON

- Engages or disengages the Autothrottle

5. SPEED SELECTOR KNOB

- Used to select the speed for the autothrottle
- Press in the middle to toggle between KIAS/MACH in manual speeds
- If FMS Speed is active, the MACH/KIAS change happens automatically

6. FMS SPEED SWITCH

- **MAN:** Selected airspeed readout is controlled manually
- **FMS:** Selected airspeed readout is controlled by the FMS logic
 - See FMS SPEED chapter for more information
 - Setting the switch to FMS speed is only possible if a flight plan has been loaded and activated in the FMS.

7. APP BUTTON

- Arms, activates, or deactivates approach modes based on the navigation source displayed on the PFD
- The following lateral and vertical mode annunciations are displayed on the FMA after the APP button is pressed:
 - LOC and GS for the ILS Approach mode
 - LNAV and GP for the VGP approach mode (This approach is not currently simulated)

Note: If you have PREV active and you pressed APP to arm an ILS approach, the autopilot source will automatically switch from FMS to V/L once the localizer deviation is less than 1 dot.

8. BANK BUTTON

- Enables and disables bank angle limit of 17° used by the Heading mode.

VERTICAL GUIDANCE CONTROLS



1. FLCH BUTTON

- Selects and deselects the Flight Level Change mode.

2. VNAV BUTTON

- On the ground: Button arms/disarms VNAV
- In the air: Selects and deselects the VNAV mode (FMS vertical navigation).

3. ALT BUTTON

- Selects and deselects the altitude holding mode (ALT).

4. ALT SELECTOR KNOB

- Selects the desired altitude.
- Pressing this knob displays on the PFD the following altitudes, in meters.
 - The selected altitude, above the selected altitude readout window
 - The actual airplane altitude, above the altitude rolling digits readout window.

5. FPA BUTTON

- Selects and deselects the Flight Path Angle mode.
- Pilot selectable pitch altitude on degrees of climb or descent.

6. FPA SELECTOR KNOB

- Manually selects the desired Flight Path Angle.

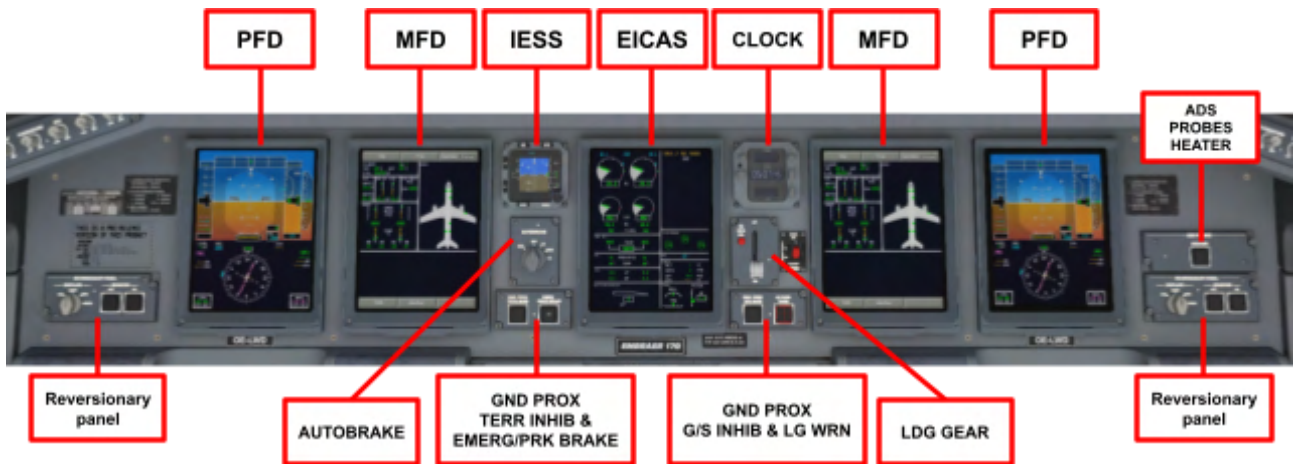
7. VS BUTTON

- Selects and deselects the Vertical Speed mode

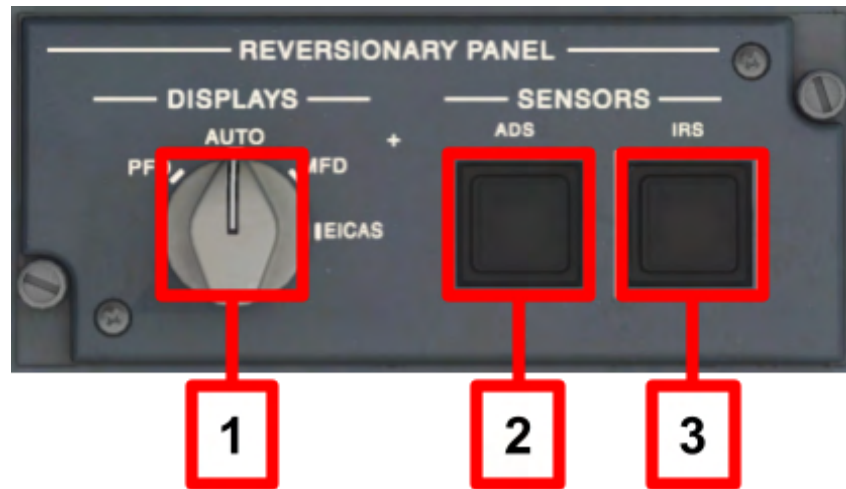
8. VS THUMB WHEEL SELECTOR

- The thumb wheel selector manually selects the desired vertical speed rate.

MAIN PANEL



REVERSIONARY PANEL



1. DISPLAY SELECTOR KNOB

- PFD: Displays PFD information on the associated MFD
- AUTO: Automatically reverts the MFD in case of display failure (Not currently simulated as we cannot simulate display failures)
- MFD: Displays MFD information on the associated MFD
- EICAS: Displays EICAS information on the associated MFD

2. ADS SENSOR BUTTON (Momentary action)

- **Note:** This function is not simulated in this version of the product

3. IRS SENSOR BUTTON (Momentary action)

- **Note:** This function is not simulated in this version of the product

PRIMARY FLIGHT DISPLAY (PFD)



AVAILABLE AVIONICS VERSIONS

- You can change the PFD avionics version on the EFB Settings page.
- This is only a visual change of the background, all systems and settings were developed based on Load 27.

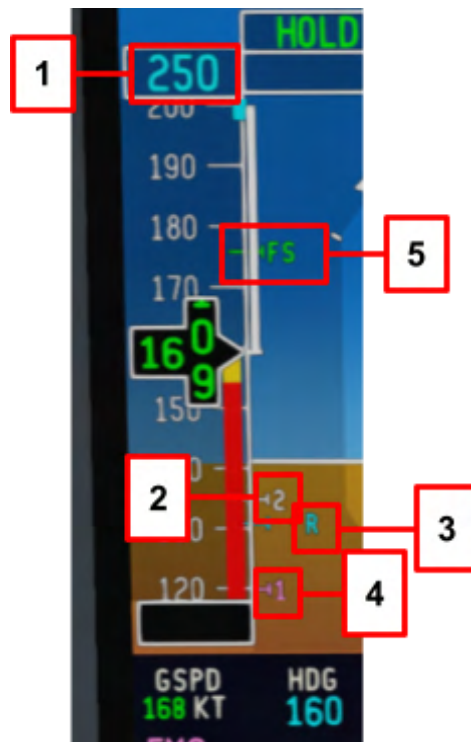


Avionics Load 27



Avionics Load 25

AIRSPEED READOUT



1. SELECTED AIRSPEED READOUT

- Displays speed/mach values
- **MAGENTA**: Automatically set by the FMS
- **CYAN**: Manually entered by the pilot

2. V_2 SPEED (2)

- Takeoff safety speed

3. V_R SPEED (R)

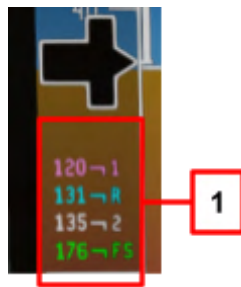
- Takeoff rotation speed

4. V_1 SPEED (1)

- Takeoff decision speed

5. V_{FS} SPEED (FS)

- Takeoff final segment speed



1. V_{SPEED} BUGS

- Preview mode displays in the lower portion of the airspeed tape. When no valid data is available and the aircraft is on the ground, cyan dashes are displayed.

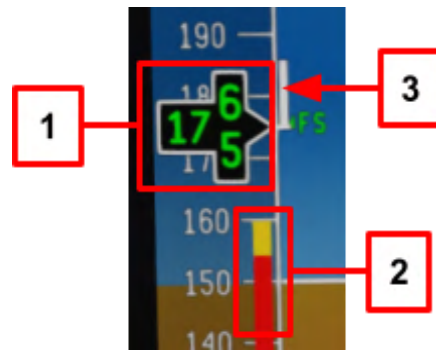
2. LOADING V_{SPEEDs}

- V_{SPEEDs} are entered in the MCDU manually, or automatically via the EFB
 - A. Go to the EFB > Takeoff perf > Fill out all the fields based on the weights calculated on the EFB and weather/airport conditions > Press "Compute". After the Vs speeds have been computed, you can send them to the FMS by pressing "**Send to FMS**" in the top right corner.
 - B. **Authentic FMS:** Press the DLK (datalink) button. You will now see the landing speeds option highlighter green. If you press the right soft key next to the highlighted text, you will load the vs speeds

Note: You can use the same process for landing speeds from the Landing perf page on the EFB.

AIRSPEED TRANSITION

- Airspeed transitions automatically between IAS and Mach and vice versa at approx 29,000ft.



1. AIRSPEED ROLLING DIGITS

- GREEN: Normal range
- AMBER: Airspeed is at the amber low speed awareness tape
- RED INVERSE VIDEO: Airspeed is at the red low speed awareness tape or at the V_{MO}/M_{MO} barber pole

2. LOW SPEED AWARENESS (LSA) TAPE

- Indicates the margin to the stick shaker speed. It rises from the bottom of the speed tape and has two colored ranges.

3. AIRSPEED TREND VECTOR

- White thermometer starting at the airspeed reference line and extends to the airspeed the aircraft is at in 10 seconds, based on current acceleration.



1. V_{MO}/M_{MO} BARBER POLE

- Displayed when the V_{MO}/M_{MO} is within the viewable range, and covers speeds at or above V_{MO}/M_{MO}
- V_{MO}/M_{MO} redline on PFD is 280 knots up to 8000ft, then increases linearly to 320 knots at 10,000' until mach changeover and then decreases.
- Overspeed Barber pole is also affected by the flaps/slats positions:
 - i. Flaps 1..... 230 KIAS
 - ii. Flaps 2..... 215 KIAS
 - iii. Flaps 3..... 200 KIAS
 - iv. Flaps 4..... 180 KIAS
 - v. Flaps 5..... 180 KIAS
 - vi. Flaps Full..... 165 KIAS

○

2. SELECTED AIRSPEED BUG

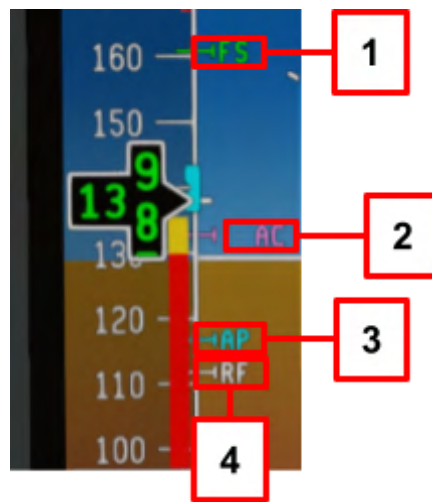
- **MAGENTA**: Displayed at the FMS airspeed target when FMS mode is valid.
- **CYAN**: Displayed at the MAnual mode airspeed target.

3. MACH AIRSPEED DIGITAL DISPLAY

- Mach digital display

4. GROUNDSPPEED

- Displayed in the upper left corner of the HSI just below the airspeed tape.



1. V_{FS} SPEED (FS)

- Takeoff final segment speed

2. V_{AC} SPEED (AC)

- Approach climb speed

3. V_{AP} SPEED (AP)

- Approach speed

4. V_{RF} SPEED (RF)

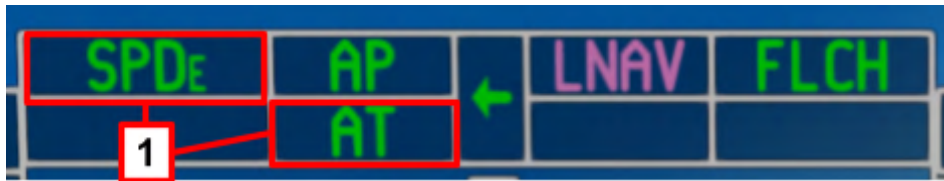
- Reference speed

FLIGHT MODE ANNUNCIATION DISPLAY (FMA)

FMA AND PFD SOURCE MODES

- **MAGENTA**: AP modes that are using FMS data
- **GREEN**: Manual AP modes

AUTOTHROTTLE ANNUNCIATIONS ON FMA



LEFT COLUMN

- The top row indicates autothrottle engagement mode, bottom row indicates autothrottle armed mode.

RIGHT COLUMN

- Depicts whether the Autothrottle is engaged or disengaged

AUTOTHROTTLE (AT) MODE FIELD ANNUNCIATORS		
Annunciator	Status	Mode Description
TO	Armed	AT Take-off Mode Armed
TO	Active	AT Take-off Mode Active
HOLD	Active	AT HOLD Active
SPD _T	Active	AT Speed on Throttle Mode
SPD _E	Active	AT Speed on Elevator Mode
RETD	Armed	AT Retard Mode
RETD	Active	AT Retard Mode
GA	Active	AT GO-Around Mode
LIM	Active	AT Limit Mode

AT	Engaged	AT is Engaged
AT	Disengaged	AT has been disengaged
OVRD	Active	AT Override is Active

AUTOPILOT ANNUNCIATIONS ON FMA



1. AUTOPILOT ENGAGE ANNUNCIATOR

AUTOPILOT MODE FIELD ANNUNCIATORS		
Annunciator	Status	Mode Description
AP	Engaged	AP is Engaged
AP	Disengaged	AP is Disengaged (Flashes in Red)

2. AUTOPILOT/AUTOTHROTTLE SOURCE SELECTION

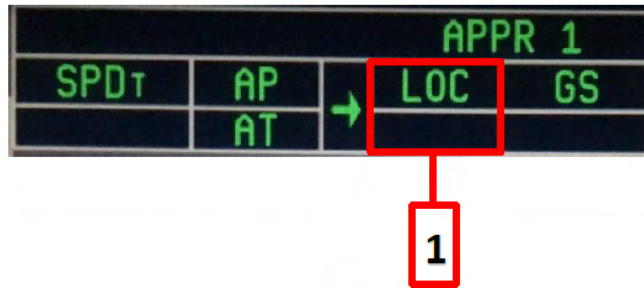
- PFD source that currently supplies the references (primary NAV source and selected airspeed reference) for the AP
- **Note:** This function is not simulated in this version of the product

3. APPROACH STATUS INDICATION

- Approach armed status shows to the left side of the source selector in white
- Approach engaged status is shown to the right of the source selector in green
- **APPR 1** (CAT I) will be Engaged and green if the following conditions are met:
 - i. Minimums are set to BARO
 - ii. Flaps are set to 5 or FULL
 - iii. Radio altitude is less than 1500ft
- **APPR 2** (CAT II) will be Engaged and green if the following conditions are met:
 - i. Minimums are set to RA
 - ii. Flaps are set to 5 or FULL

- iii. Radio altitude is less than 1500ft
- **L/VNAV** will be shown for RNAV approaches with vertical guidance
- **LPV** will be shown for RNAV approaches with SBAS (Satellite Based Augmentation System)

LATERAL FLIGHT DIRECTOR MODES



LATERAL MODE FIELD ANNUNCIATORS		
Annunciator	Status	Description
Armed Modes (Bottom row)		
LOC	Armed	Localizer Mode
LNAV	Armed	Lateral Navigation Mode
Active Modes (Top Row)		
ROLL	Active	Roll Hold Mode
HDC	Active	Heading Hold Mode
TRACK	Active	Take Off or GO-Around Mode
LOC	Active	Localizer Mode
LNAV	Active	Lateral Navigation Mode

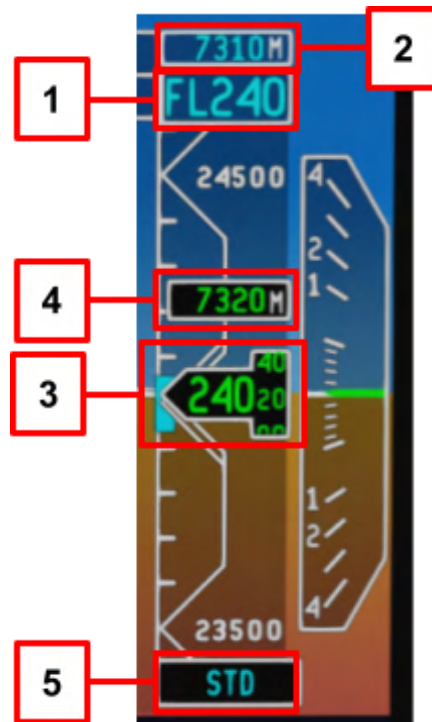
VERTICAL FLIGHT DIRECTOR MODES



VERTICAL MODE FIELD ANNUNCIATORS		
Annunciator	Status	Description
Armed Modes (Bottom row)		
GS	Armed	Glideslope Mode
GP	Armed	Glidepath Mode
VNAV	Armed	Vertical NAV Mode
Active Modes (Top Row)		
GS	Active	Glideslope Mode
ALT	Active	Altitude Hold Mode
FLCH	Active	Flight Level Change Mode
TO	Active	Takeoff Mode
FPA	Active	Flight Path Angle Mode
ASEL	Active	Altitude Select Mode
VS	Active	Vertical Speed Mode
VNAV	Active	Vertical NAV Mode
FLCH	Active	Flight Level Change Mode (From FMS)
ASEL	Active	Altitude Select Mode (From FMS)
ALT	Active	Altitude Hold Mode (From FMS)
PTH	Active	Path Mode and Vertical PAtH (From FMS)

GP	Active	Glidepath Mode
OVSP	Active	Overspeed Protection is Active

ALTITUDE READOUT



1. SELECTED ALTITUDE DIGITAL READOUT

- Displays altitude rounded to 100ft
- Format is in thousands of feet or FL, depending on the STD setting. More below in number 5.
- Displays cyan dashes on ground power-up. These disappear once you rotate the altitude knob for the first time
- Altitude alert sounds when the aircraft is within 1000 ft of selected altitude

2. METRIC SELECTED ALTITUDE READOUT

- This readout shows the preselected altitude in meters.
- It can be displayed or removed by the PUSH FT-M button that is in the middle of the ALT knob

3. ALTITUDE ROLLING DIGITS DISPLAY

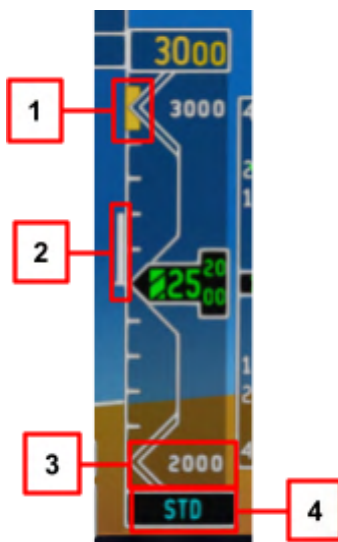
- Actual altitude rolled to a resolution of 1 foot, but labeled every 20 feet.
- Altitudes above 0 but less than 10,000 ft have a green black dashed box symbol that replaces the ten thousand digit

4. METRIC ALTITUDE DISPLAY

- Current altitude in meters is displayed to a resolution of 5 meters.

5. BARO ALTIMETER SETTING

- Value is adjusted with BARO SET knob on the Guidance panel
- When the PUSH STD button on the BARO SET knob is pushed
 - i. The altimeter setting window changes to STD
 - ii. The correction is set to 29.92 in Hg or 1013 hPA
 - iii. Dialed altitude is shown in the Flight Level Format. (Example: "FL210" instead of "21 000")
- Selected altitude digital readout then displays Flight Level (e.g. "FL300" instead of 30,000ft)



1. SELECTED ALTITUDE BUG

- Corresponds to the altitude value set in the altitude alert select digital display.

- When the selected altitude is off the displayed range, half of the bug remains on the top or bottom of the scale to indicate the direction to the selected altitude.

2. ALTITUDE TREND VECTOR

- White thermometer starting at the altitude reference line and extends to the altitude the aircraft will be at in 6 seconds, based on current vertical speed.

3. BAROMETRIC ALTITUDE TAPE

- Moving scale.
- White tick marks are at 100-foot increments, except the white chevrons are at the 500-foot increments. Double chevrons are displayed at the 1,000-foot increments.

4. BARO (BAROMETRIC) ALTIMETER SETTING

- Value is adjusted with BARO SET knob on the guidance panel.
- When the PUSH STD button on the BARO SET knob is pushed the altimeter setting window changes to STD and the correction is set to 29.92 in Hg or 1013 hPa.

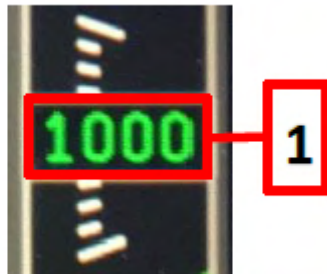


1. SELECTED VERTICAL SPEED READOUT and BUG

- Displayed when VS is the active vertical flight director mode.
- Vertical speed is manually entered by using the VS when AP is in VS mode.
- Cyan arrows above or below the display indicate the aircraft must climb or descend to attain the target vertical speed.

2. VERTICAL SPEED SCALE

- Fixed scale with moving green pointer.
- Green pointer indicates the vertical speed on the scale.
- Cyan filled bug is displayed on the vertical speed scale and is positioned at the selected vertical speed readout value.



1. VERTICAL SPEED DIGITAL READOUT

- Displays digital vertical speed.
- If TCAS RA is active, vertical speed readout is removed.
- At vertical speeds greater than 9,900 FPM, when the pressure altitude data received is invalid, or when the altitude rate data is invalid the readout displays:

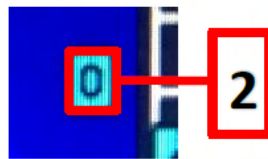


1. RADIO ALTITUDE READOUT

- Only displayed when radio altitude is valid and less than 2,500 feet.

2. SELECTED MINIMUM RA/BARO ALTITUDE READOUT AND BUG

- Set using the MINIMUMS rotary knob when BARO is selected on the MINIMUMS RA/BARO select switch.
- If additional counterclockwise clicks occur after the display reads 0, display is removed from view.
- **B** Selected BARO altitude bug (with a horizontal L that sticks out from the left edge of the altitude tape).
- **R** Selected Radio Altitude bug (with a horizontal L that sticks out from the left edge of the altitude tape).



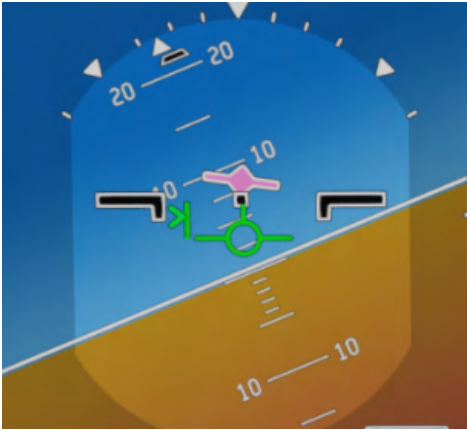
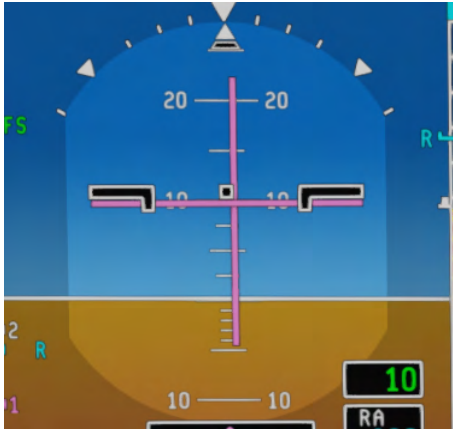
2. MARKER BEACONS

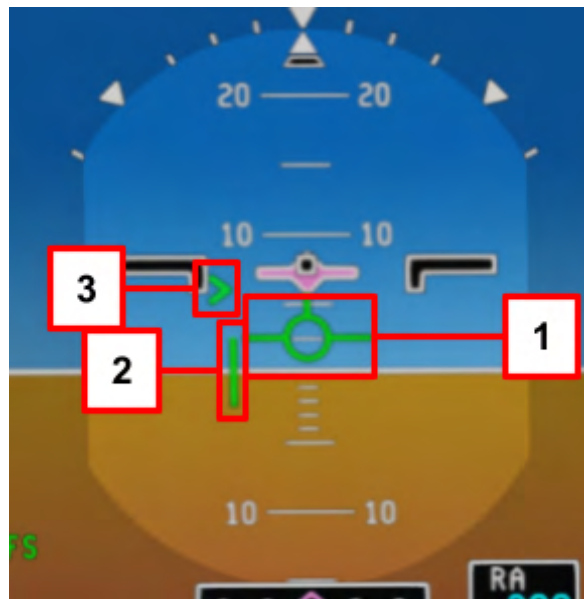
- Located in the upper right corner

Annunciator	Marker	Color
O	Outer	Cyan
M	Middle	Yellow
I	Inner	White

FLIGHT DIRECTOR (FD)

- Automatically displays if either is true:
 - TOGA button actuation
 - Autopilot is activated
- FD automatically transitions from TAKEOFF MODE to GUIDANCE CUE upon reaching 1200ft after Takeoff.

FLIGHT DIRECTOR GUIDANCE CUE (Diamond with short wings)	FLIGHT DIRECTOR GUIDANCE CUE IN TAKEOFF MODE (Cross bars)
• Displayed as a magenta diamond with short wings. • Fits within the flight path vector symbol. 	• Magenta vertical and horizontal bar • Displayed only when TOGA is pressed on the throttle which activates the TO flight director mode • Maintains extended runway centerline. • Maintains pitch attitude based on appropriate configuration. 



1. FLIGHT PATH VECTOR (FPV)

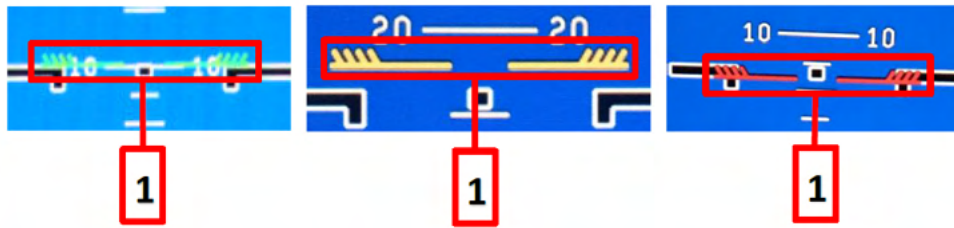
- Consists of a green aircraft symbol that indicates the current flight path angle in reference to the horizon line.
- FPV is removed when excessive pitch is exceeded.

2. FLIGHT PATH VECTOR SPEED ERROR TAPE

- Green tape that rises or descends from the left wing of the FPA symbol.
- Represents the difference between the selected airspeed and the current airspeed.
- Down indicates current airspeed is less than the selected airspeed, and up indicates more.

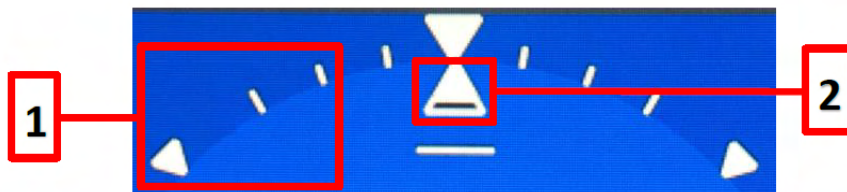
3. FLIGHT PATH VECTOR ACCELERATION POINTER

- Displayed when FD is selected and gives an analog indication of acceleration and deceleration rate along the flight path.
- Moves up for increasing values of flight path acceleration and moves down for decreasing values of flight path acceleration.



1. PITCH LIMIT INDICATOR (PLI)

- Pitch-based indication between stick shaker angle of attack and airplane's angle of attack.
- PLI is displayed as Green, Amber, or Red.
- Red PLI - stick shaker is activated.



1. ROLL SCALE

- Tick marks are located every 10° up to 30°.
- Inverted triangles are displayed at 0° and 45° of roll.

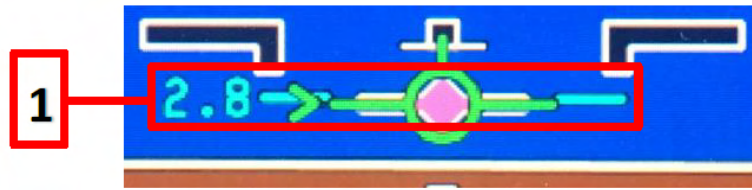
2. ROLL POINTER AND SLIP/SKID INDICATION

- The triangle is the roll pointer. Bottom half of the roll pointer is used as a slip/skid indicator.
- Lower portion of the triangle moves left or right to show the slip or skid condition.



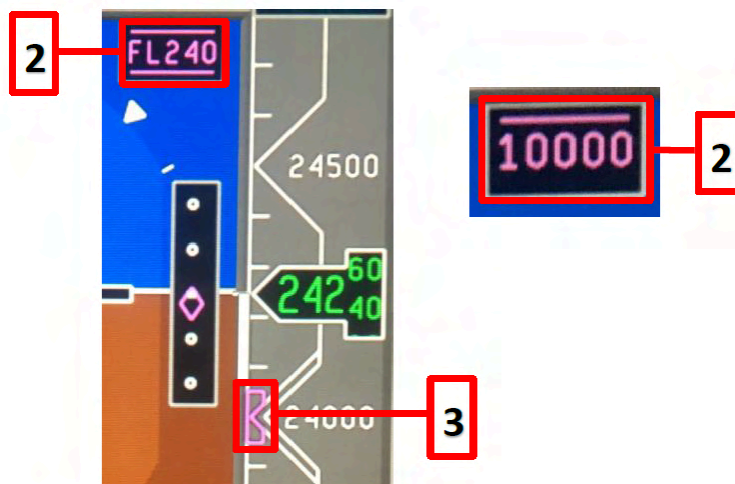
3. LOW BANK LIMIT ARC

- Helps the pilot to fly with low bank angles during turns
- Automatically displayed above FL250



1. FLIGHT PATH ANGLE (FPA)

- indicates the current flight path in reference to the horizon line.



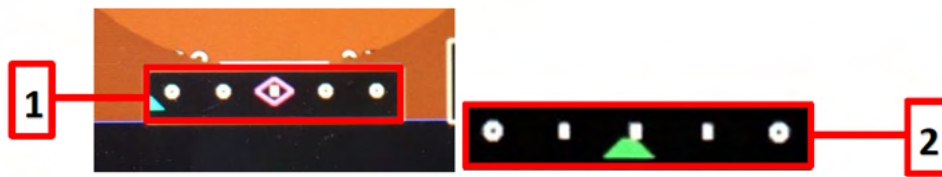
2. VNAV ALTITUDE WAYPOINT CONSTRAINT READOUT

- Displays an active waypoint constraint altitude automatically honored by the FMS prior to arrival at the preselected altitude.

Constraint Type and Phase of Flight	Display Method
AT	<u>XXXXX</u> over-scored and under-scored
AT OR ABOVE	<u>XXXXX</u> under-scored
AT OR BELOW	<u>XXXXX</u> over-scored

3. VNAV TARGET ALTITUDE BUG

- Displayed when a VNAV target altitude is valid and the FMS is the primary NAV source.

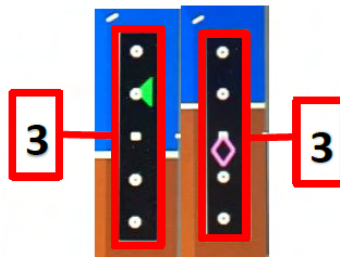


1. ADI LATERAL DEVIATION

- Magenta diamond for FMS deviation. Green trapezoid pointer used for LOC.
- When both are displayed during PREV mode the trapezoid is displayed in cyan.
- When a valid deviation is not available, a **X** is depicted.

2. EXPANDED LATERAL DEVIATION

- Replaces the ADI scale for CAT II operations.
- Higher lateral deviation sensitivity.



3. VERTICAL DEVIATION SCALE AND POINTER

- Pointer is controlled by the selected primary navigation source (PNS).
- Vertical deviation scale is displayed when:
 - Localizer (LOC) is selected as the PNS.
 - FMS is selected as the PNS, and the FMS commands the VNAV vertical scale in view.
- ILS LOC deviation pointer is a green trapezoid.
- VNAV FMS is a magenta hollow diamond
- When glideslope data is invalid or when the vertical deviation scale is commanded into view by the FMS and a valid deviation is not available a **X** is depicted.
- Preview vertical deviation is displayed when LOC is the PREV source.
- LOC and FMS deviation are displayed the same way as the lateral deviation.



FULL COMPASS MODE



ARC MODE

HORIZONTAL SITUATION INDICATOR (HSI) DISPLAYS

- Heading display on the HSI has 3 selectable modes: Full compass, Arc, and Map mode.
- Modes are selected by pushing the HSI button.

FULL COMPASS MODE

- Displayed on initialization.
- 360-degree compass view.
- Present heading is indicated by a fixed lubber line at the top of the heading scale with a digital heading readout directly above the lubber line.

ARC MODE

- 90 degrees of compass view.
- Weather is displayed when the radar is on and the **WX** button next to the HSI button is pushed





1. SELECTED HEADING DIGITAL READOUT AND BUG

- Displays the digital value for the aircraft selected heading.
- Cyan heading select bug is positioned by rotating the HDG SEL knob.
- Pushing the PUSH SYNC button on the HDG SEL knob synchronizes the heading bug and readout to the current aircraft heading.

2. DRIFT ANGLE BUG

- Represented by an inverted solid white triangle displayed on the outside edge of the heading scale.
- The difference between the aircraft heading and the aircraft ground track is the drift angle.
- The drift angle bug is placed on the edge of the heading scale and aligned to the drift angle with respect to the lubber line.
- When the drift angle is 0°, the drift bug fits inside the heading bracket notch.

3. WIND DISPLAY

- Digital readout for wind speed and an arrow vector for direction.



1. PRIMARY NAV SOURCE ANNUNCIATOR

- Selected using the V/L or FMS buttons.
- The **V/L (VOR/LOC)** button selects the available SRN (short-range navigation) source, displayed in green.
 - The associated annunciators are **VOR 1** **VOR 2** **LOC 1** **LOC 2**
- The **FMS** button selects FMS as the source, displayed in magenta.
 - With dual FMS systems installed the identifier will show **FMS1** or **FMS2** (this feature is not currently implemented)

2. PRIMARY NAV DISTANCE IDENTIFIER

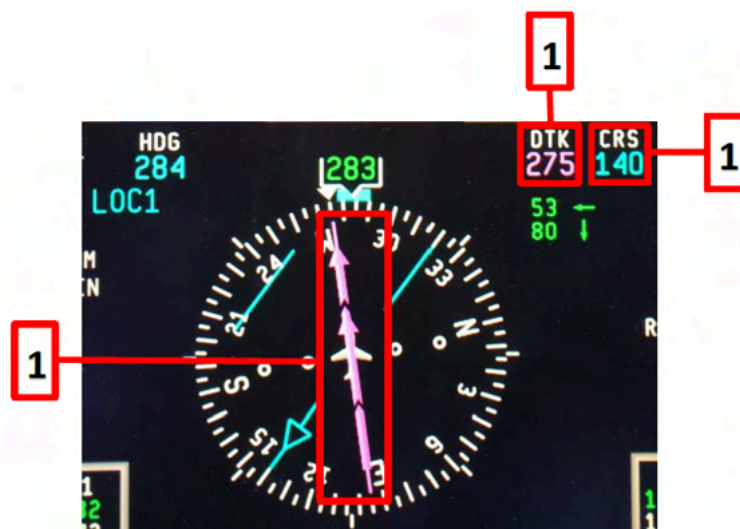
- Displays the NAV source station identifier.
- The next magenta TO waypoint identifier is displayed.

3. PRIMARY NAV DISTANCE READOUT

- Distances of 100 NM or more, the digits are displayed with 1 NM resolution.
- Distances of less than 100 NM, the digits are displayed with 0.1 NM resolution.

4. PRIMARY NAV ESTIMATED TIME ENROUTE

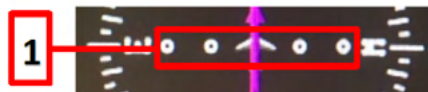
- Time is always displayed in minutes.



1. COURSE SELECT/DESIRED TRACK POINTER AND READOUT

- Green for VOR and LOC navigation and the label is **CRS**
 - When VOR or LOC is selected, the track pointer is positioned by turning the CRS knob.
- Magenta for FMS navigation and the label is **DTK**

- Value of the desired track is supplied by the FMS.
- Course deviation bar moves laterally and shows lateral deviation.



1. COURSE DEVIATION INDICATOR (CDI) AND LATERAL DEVIATION SCALE

- Only displayed when a navigation source is active.
- Deviation bar moves left or right to show the deviation from the selected NAV source.



2. BEARING POINTERS

- Source is either VOR, ADF, or FMS.
- Source is selected using the **BRG** ○ or **BRG** ▼ buttons on the guidance panel



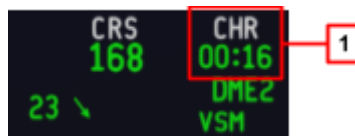
1. DME SOURCE ANNUNCIATOR

- DME source is selected by pushing the V/L button.
- DME source annunciation reflects the short-range navigation source selection.

- VOR1/LOC1 displays **DME1** VOR2/LOC2 displays **DME2**
- When the primary NAV source is FMS, the last selected VOR/LOC is used for the DME distance display.
- When the same source is used, the DME data is displayed in amber.

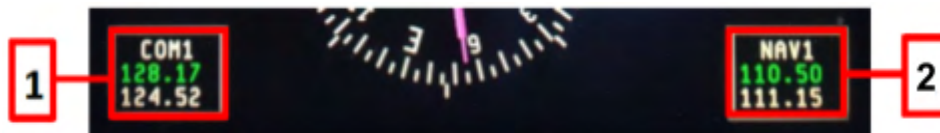
2. DME STATION IDENTIFIER, DISTANCE, AND TTG READOUT

- DME source station identifier is displayed directly below the DME source annunciator.
- Distance to go to the waypoint identified by the DME station identifier is shown in nautical miles.
- TTG to the waypoint identified by the DME station identifier is shown in minutes.



1. ELAPSED TIMER

- Located in the top right corner of the HSI.
- Displays time in hours and minutes (HH:MM) or minutes and seconds (MM:SS).
- Controlled by the CHRONO button on the control yoke.
- 1st push - displays readout with the readout counting up from 00:00. Next push pauses the readout. Next push resets the readout back to 00:00 and pauses. Next push restarts the count.



1. COM FREQUENCY AND IDENTIFIER DISPLAY

- White COM source identifier annunciator.
- Green COM active frequency active frequency.
- White COM standby frequency

2. NAV FREQUENCY AND IDENTIFIER DISPLAY

- White NAV source identifier annunciator
- **Green** frequency indicates the active **manually** tuned frequency
- **Magenta** frequency indicates the active frequency **auto-tuned** by the FMS
- **White** NAV standby frequency.

In other words, if the active frequency is magenta, then ILS Autotune is active. If the active frequency is green then ILS Autotune is not active.

PREVIEW MODE

- When FMS is the primary navigation source and preview is selected, course, lateral deviation and vertical deviation are displayed.
- This is usually used when nearing the approach, prior to switching to VOR/LOC for the approach.
- Only available in full compass or arc mode.

PREVIEW MODE ACTIVATION

- The PREVIEW button can be turned on manually by pressing the PREV button on the glareshield panel.
- If ILS Autotune is active, PREV will be displayed automatically as soon as the ILS frequency has been tuned.
- This happens 150NM out of the airport when the aircraft enters the terminal (TERM) area.
- If the total flight plan is shorter than 150NM, then the autotuning will happen at 50NM from the destination airport.



1. PREVIEW NAVIGATION SOURCE

- Preview NAV source is annunciated in cyan to the right of the primary NAV source.

- Available annunciations are: **VOR1** **VOR2** **LOC1** **LOC2**.

2. PREVIEW COURSE POINTER, DEVIATION BAR, and DIGITAL READOUT

- Cyan course arrow and deviation bar displayed at the same time as the primary NAV source DTK pointer and deviation bar.
- Changed by turning the on-side CRS select knob when PREV is displayed.
- TO/FROM pointer is a smaller cyan arrowhead that is displayed at the top (TO) or bottom (FROM) of the deviation bar.

TERM & APPR ANNUNCIATORS

- TERM:**

- 30NM out of the destination airport the aircraft enters the terminal area depicted by the TERM annunciator on the lower right of the PFD.
- This moment indicates when the APPROACH SPEEDS can be activated on the FMS if FMS Speed is used, or and when the ILS Autotune features tunes the approach frequency.



- APPR:**

- Indicated when we are 2NM away from the Initial Approach Fix, or 5NM from the runway threshold, whichever comes sooner.

MSG ANNUNCIATOR

- This annunciator indicates that there is a message on the FMS Scratchpad that the pilot should pay attention to.
- MSG flashes in reverse video until the message on the scratchpad has been cleared.
- **Tip:** You can clear the message quickly by pressing TAB-BACKSPACE-TAB on your keyboard, which will activate the Direct Keyboard Entry, clear the message with backspace and deactivate DKE again.



ILS (FMS) AUTOTUNE

OVERVIEW

The ILS autotune feature automatically loads the correct NAV frequency for the approach you selected via the FMS and it automatically switches from LNAV to LOC at the right moment.

This means that the pilot does not have to manually tune in the ILS frequency or switch from FMS to the LOC source manually.

TUNING THE FREQUENCY

- The Autotune feature will automatically load the correct NAV 1 or NAV 2 frequency into the radios based on the approach selected in the FMS.
- This happens 150NM out of the airport when the aircraft enters the terminal (TERM) area.
- If the total flight plan is shorter than 150NM, then the autotuning will happen at 50NM from the destination airport.
- There are two places where you can verify that ILS autotune is currently **active**:
 - The NAV FREQUENCY AND IDENTIFIER DISPLAY on the PFD displays the active frequency in **magenta**



- The FMS RADIO page displays “FMS AUTO” is displayed on the FMS RADIO page.



HOW TO DEACTIVATE ILS AUTOTUNE

Manually entering or changing the active NAV frequencies will automatically deactivate ILS autotune.

HOW TO ACTIVATE ILS AUTOTUNE

Autotune is on by default. If you deactivated ILS autotune and you would like to reactivate it again, you can do so by clicking on the white standby frequency two times, first to select it, and the second time to confirm the selection which will bring you to a new NAV1 / NAV2 page.

Here you can turn FMS AUTO back ON:



AUTOMATIC SWITCH FROM FMS TO LOC

- The ILS AUTOTUNE includes a feature which will automatically switch from **FMS** as the source of the autopilot (The AP is following the route from the FMS in LNAV) to **V/L** (Autopilot follows the localizer of the approach - LOC).
- Switching will disengage VNAV and engage FPA as the vertical mode, with GS armed.

The automatic switch happens if the following conditions are met:

1. PREV 1 or PREV 2 have been selected by the pilot.
 - Selecting PREV 1 will then switch to NAV 1
 - Selecting PREV 2 will then switch to NAV 2
2. The approach has been armed by the pilot by pressing the APP button
3. The lateral deviation of the Localizer in preview mode is less than one dot

HOW TO FLY RNAV APPROACHES

The E-Jets Family 2.0 can correctly fly all RNAV approaches, with a few exceptions where the aircraft will still complete the approach but with minor deviations.

For any approach with vertical guidance—whether **ILS or RNAV**—simply press the **APP** button. The autopilot will then activate the appropriate vertical guidance:

- **For ILS approaches**, this will be the **Glide Slope**.
- **For RNAV approaches**, this will be the **Glide Path**, which functions exactly like the Glide Slope but is displayed with magenta indications.
 - On the PFD, you will see either **L/VNAV** or **LPV** displayed depending on what kind of RNAV approach you're flying. For any approach that includes the WAAS/EGNOS channel on the charts, you'll see LPV. For any other RNAV approach, you'll see L/VNAV.

The only limitation involves **lateral guidance** for RNAV approaches that include curved legs with flyover waypoints. In these cases, the aircraft will fly the segments **point to point**, causing the autopilot to cut corners slightly. However, the system will still guide you accurately to the runway for all approaches.

MULTIFUNCTION DISPLAY (MFD)



1. TITLE MENU BUTTONS (HOT KEYS)

- Each have menus to control and show the associated MFD display pages.
- Controlled using the Cursor Control Device (CCD)
- Note:** TCAS and Weather submenus are not simulated in the current version. These sub menus contain additional settings for TCAS and weather radar which are not available to us in the simulator. Otherwise, TCAS and Weather overlays are triggered through the MAP or PLAN dropdown menu.

2. PRIMARY NAV SOURCE, TO WAYPOINT IDENTIFIER, DISTANCE, and TIME DISPLAY

- Displays the primary NAV source on the top row, waypoint identifier, Distance readout below and the bottom line contains the time-to-go readout (If the aircraft is not moving, time-to-go is not displayed).

3. WIND DISPLAY, CARTESIAN

- Cartesian wind display displays two vector arrows.

4. FLIGHT INFORMATION DISPLAY

- Displays Static Air Temperature (SAT), Total Air Temperature (TAT), and True Airspeed (TAS).
- Use SAT when ATC asks for OAT.
- SAT should be used when entering temperatures in the EFB Performance pages.

5. DME DISPLAY

- Displays DME source, Station Identifier on the middle row, and distance readout on the bottom row.

LOWER MFD SECTION



1. PROGRESS DISPLAY

- Displays waypoint name, destination, distance-to-go, ETA/ETE, and fuel remaining based on calculations received from the FMS.

2. VERTICAL SCALE TAPE

- Used as an altitude reference for the pilot.

3. SELECTED ALTITUDE INDICATOR

- Shows the selected altitude set by the altitude select knob.

4. HALF-AND FULL-RANGE INDICATORS

- Vertical scaling ranges are selected based on the horizontal half-range currently shown.
- When range is controllable it is displayed in cyan. When the range is not controllable it is displayed in white.

5. AIRCRAFT TRAJECTORY LINE

- Shows the vector trend of the aircraft in the vertical axis based on the current flight path angle.

6. WAYPOINT DISPLAYS

- Contains a waypoint symbol, waypoint identifier, waypoint altitude, and when required a hold annunciator.

7. VERTICAL TRACK LINES

- Connect continuous waypoints in the flight plan.

8. AIRCRAFT SYMBOL

- Gives a horizontal and vertical reference point for downpath waypoints.

FLIGHT PLAN EDITING VISUALIZATION

When editing a flight plan, for example when you are adding new waypoints, the FMS goes into the MOD mode.

- In this mode, any new route segments will be displayed in cyan on the FMS, and they will be visualized with dashes, until the new flight plane is ACTIVATED on the FMS.
- If you remove a waypoint, nothing will be displayed in dashes, as there isn't a new segment to be displayed.



In the example above, we added EVAXI and ABTAN after MALUG, and we can see that those two segments are dashed, and the waypoints are blue on the FMS, since we did not activate these changes yet.

MAP MODE vs PLAN MODE



MAP MODE



PLAN MODE

Function	Map Mode	Plan Mode
Display Orientation	Heading up	North up
View	Looking at map from nose of aircraft	Looking at map relative to North Up
Nonmoving reference at center of display	Airplane	Next Waypoint (Aircraft centered not implemented)
Heading indication	Compass	Aircraft symbols nose (no compass displayed)
What moves (on display)	The Map	The Aircraft
Range measured from:	Aircraft symbol	Waypoint
TCAS overlay	YES	NO
Weather overlay	YES	NO
EGPWS overlay	YES	NO
Cycle through the fplan	NO	YES

MAP ZOOM and CYCLING THROUGH THE PLAN



1. CHANGING THE MAP RANGE

- This number shows the range of the white circle (half range). In this particular case, it is 20 miles.
- **This is also a hidden manipulator!** You can use it to **zoom in or out on the map**. The official way of doing that is by rotating the knob on the pedestal under the FMS, which is also possible, but we found the manipulator on the screen much more convenient for a sim use.
- Displays waypoint name, destination, distance-to-go, ETA/ETE, and fuel remaining based on calculations received from the FMS.

2. CYCLING THROUGH THE FLIGHT PLAN IN PLAN MODE

You have several ways of cycling through the flight plan:

1. Hidden Manipulator

Highlighted with number two on the screenshot above is a hidden manipulator. When you are in the PLAN MODE, click here (left or right, or scroll with your

mouse wheel) to cycle through the flight plan.

2. **Knob on the pedestal**

Use the **inner** knob on the pedestal that is located just below the left FMS (below). This is how it is done in the real aircraft.

3. **Use your keyboard**

Assign our custom commands to any key on your keyboard or hardware in order to easily cycle through the flight plan.

Go to the X-Plane Keyboard settings, and in the command field, search for “xcrafts” and then in the FMS section, you’ll see two commands:

1. ***Center the plan map on the next waypoint***
2. ***Center the plan map on the last waypoint***

TCAS PAGE

Traffic Collision Avoidance System. If **TCAS is selected** on the map Dropdown menu and TCAS **mode is TA/RA**, traffic will be displayed on the MFD.

- If Vertical Situation Display is displayed, traffic will be displayed **on the map**
- If Vertical Situation Display is not displayed, Traffic will be displayed on **the lower section of the screen.**

TCAS INDICATIONS:

- RESOLUTION ADVISORY (RA) INDICATION: Red solid square.
- TRAFFIC ADVISORY (TA) INDICATION: Amber solid circle.
- PROXIMATE TRAFFIC INDICATION: Cyan solid diamond.
- OTHER TRAFFIC INDICATION: Cyan hollow diamond.

TCAS MODE

- Annunciates the TCAS operating modes: OFF - STBY - TA/RA (ACTIVE)



TCAS Indications

TCAS Pop Up on the left of the MFD

The TCAS button on the map would just allow additional TCAS settings which we do not currently have.

WEATHER RADAR SYSTEM

If you are using 12.3.0 or newer, you will have access to the new weather radar in X-Plane.

To see the full description of the Weather Radar's capabilities, [visit this page](#).



To show the weather on the MFD, you have to:

1. Select the weather Overlay from the Map dropdown menu
2. Enable the radar through the Weather pop up menu on the bottom of the MFD

Weather Menu

Use the Weather Menu on the bottom of the MFD to turn the Radar on, and to access additional subsettings:

- **FSBY OVRD:** When you are on the ground, the radar is automatically in Forced Standby mode. **FSBY** is displayed in the mode field. The FSBY OVRD button allows you to enable radar on the ground.
- **WX:** The weather mode is displayed. In the weather mode, the system is fully operational and **WX** is displayed in the mode field.

- **GMAP:** The radar is in the ground mapping mode. In this mode, the system is fully operational, and **GMAP** is displayed in the mode field.
- **STBY:** Standby mode. Radar is energized but isn't scanning or showing any information on the MFD. **STBY** is displayed in the mode field.
- **OFF:** Radar is off. **OFF** is displayed in the mode field.
- **SECT:** The normal radar sweep is 60 degrees from the nose of the aircraft at a rate of 12 sweeps a minute. When the SECT box is selected (checked) the angle of the sweep is reduced to 30 degrees, and the rate is increased to 24 sweeps a minute. Deleting the check returns the sweep to the normal setting.
- **STAB OFF:** The radar is normally pitch and roll stabilized. The STAB OFF button disables this and slaves the radar to the nose of the plane. **STAB** is displayed in the mode field.
- **VAR GAIN:** When the VAR Gain box is not checked, the system is at the normal preset gain recommended for weather avoidance. When the VAR Gain box is checked, the system is in the variable (VAR) gain mode. **VAR** is displayed in the mode field.

Variable gain is useful for additional weather analysis. In the weather mode, variable gain increases receiver sensitivity over the calibrated level to show very weak targets, or is reduced below the calibrated level to eliminate weak returns. When the box is checked, the Gain annunciates a cyan digital value that is adjusted using the concentric knobs on the pedestal.

- **TGT** and **RCT** modes are currently not implemented since the X-Plane weather Radar doesn't include these modes.
- **ACT:** Altitude Compensated Tilt. In this mode, the system automatically tilts the antenna in relationship to the current altitude and currently selected range. When ACT is active, an **A** is displayed in the mode field next to the tilt readout.

You can also adjust the tilt manually by deselecting ACT and using the inner concentric circle of the knob on the pedestal, or use click spots around the digital readout directly on the MFD.

- **TURB:** Turbulence (TURB) -- When this box is checked, the turbulence detection mode is active. **/T** is displayed next to the **WX** in the mode field.

Weather Mode Annunciators

The current radar settings can be found in the box on the left side of the MFD towards the middle. The annunciators described above will be shown in this box. In addition, you might see **/TX**. This means that the radar is on, but you didn't enable the Weather overlay in the map dropdown menu.

EPGWS ON THE MFD (TERRAIN RADAR)

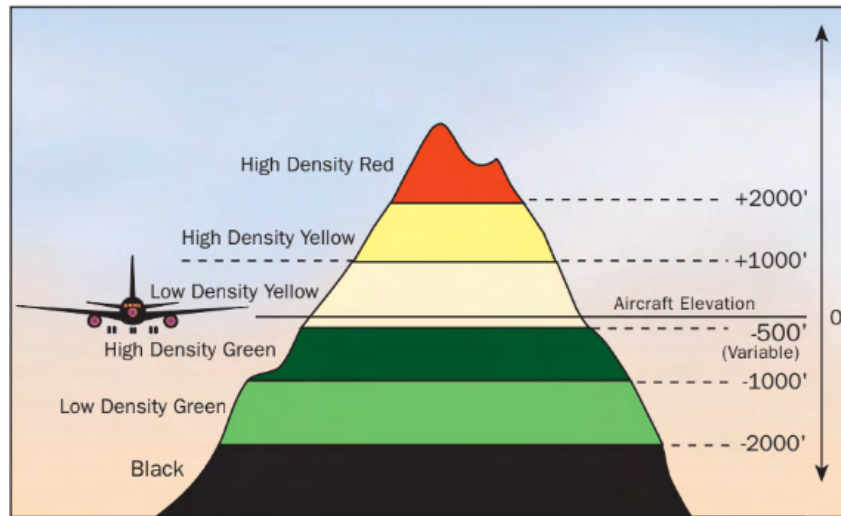
Note: This feature only works if a 3rd party plugin is installed in X-Plane: [Terrain radar + Vertical Situation Display by DrGluck](#).

The **EGPWS** is a Terrain Awareness and Alerting system providing terrain alerting and display functions with additional features.



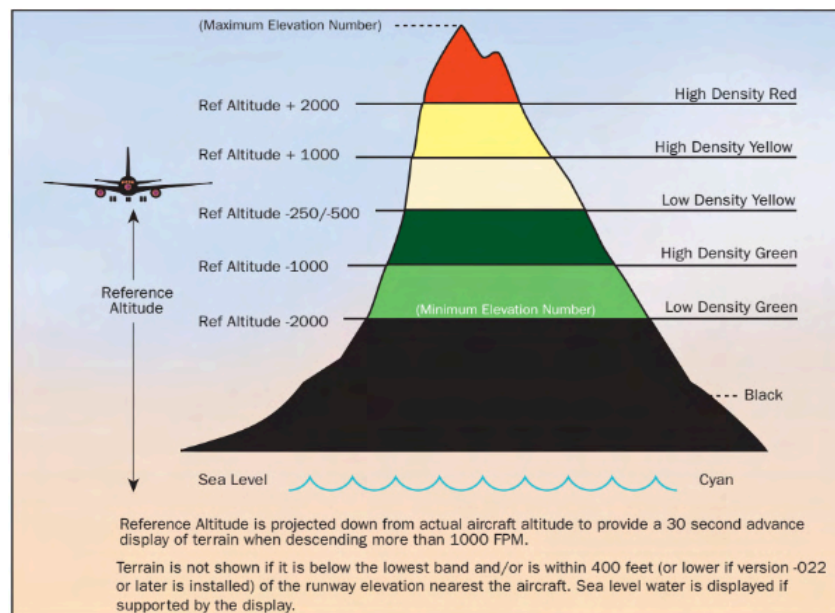
- Only displayed in the Map format
- Selected by checking the TERRAIN button on the Map dropdown menu
- The plugin can work in both modes: **Non-peaks terrain display** and **Peaks terrain display**.
- The two numbers in the TERRAIN BOX represent the upper and lower boundaries of the currently displayed terrain in hundreds of feet. If the display has a red/yellow terrain, then the upper number will be red/yellow.
- If the highest and lowest altitude numbers are the same, the lower altitude number will not be displayed.
- If the terrain is more than 2000 feet below, then the display changes to **Peaks mode**. In this mode, the terrain is displayed only in green. Display shows only the highest mountains in the area.

The **Non-Peaks display** provides a graphical plan-view image of the surrounding terrain as varying density patterns of green, yellow, and red as illustrated in the following graphics.



Peaks Display has all the characteristics of the Non-Peaks Display but with additional terrain display features for enhanced situational awareness independent of the aircraft altitude. The principle additions are:

The digital display of the highest and lowest terrain/obstacle elevations currently displayed, The display of additional solid or lower density color bands, including the addition of the graphic representation of sea level (0 feet MSL)



SYSTEMS SYNOPTIC PAGES

STATUS PAGE

- This is the only available Synoptic page of the MFD when the airplane is powered from DC power only.

Displays the following information:

- Flight info
- Temperature (TAT, SAT)
- Gross Weight
- Batteries voltage indication
- Oil Level indication
- Cockpit oxygen pressure indications
- Emergency parking brake accumulator pressure indication
- Brake temperature indication (not simulated correctly in this version)
- Doors: Green - Closed / Red - Open



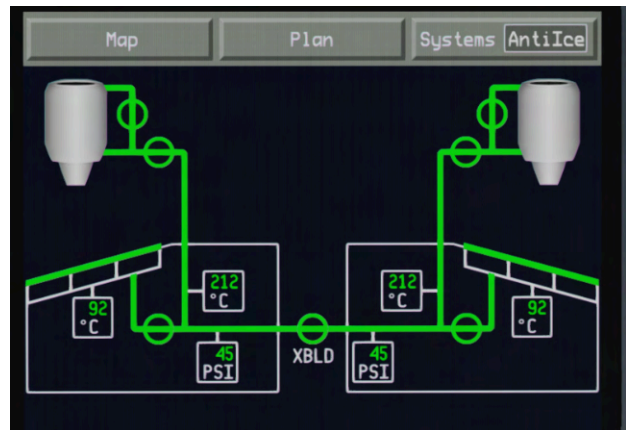
Bleed Pressure Readouts



- They display the Bleed pressure in the engine manifolds in PSI.
- Generally, pressure should be around 45PSI, but it will change depending on the APU and Engine startups.
- Air is normally tapped from the 6th stage compressor. Pushing the thrust levers up from idle to around 55% N1 causes the 10th stage to take over and supply air and you'll see the pressure drop from 45PSI to 30PSI and then increase again.
- These readouts are identical on the STATUS, ECS and ANTI-ICE pages.
- The readouts disappear from the STATUS page when engines are fully started.

ANTI-ICE PAGE

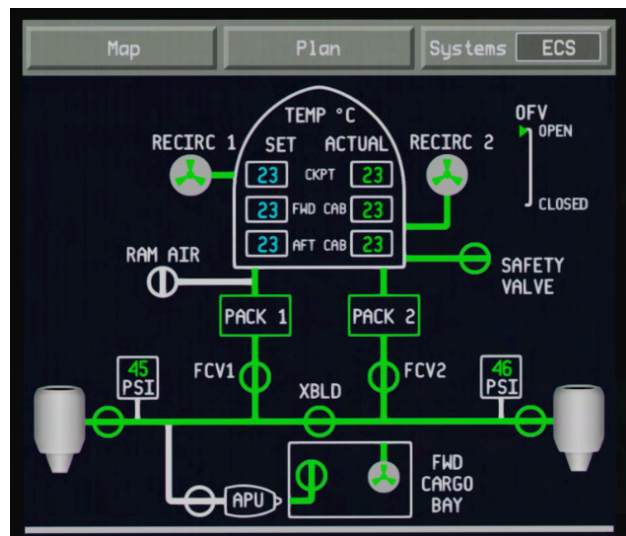
- Bleed Air System Valves Status
- Bleed Air Temperature Indication
- Bleed Air Pressure Indication
- Wing Anti-Ice Temperature
- Ice Condition Annunciation - Ice forms if flying through clouds, around 0°C. Rain or extreme cold air does not result in ice creation.



ECS PAGE

Displays the operation of the Environmental control system and contains the following elements:

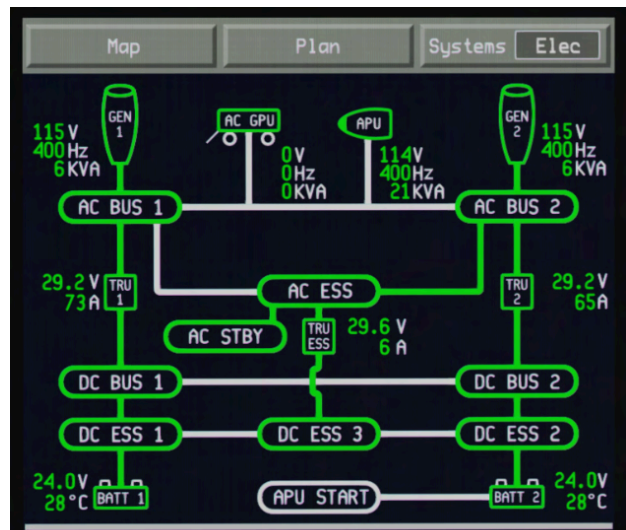
- Air shutoff valves status
- Pack Status
- Recirculation Fan Status
- Cockpit/Cabin Temperature Indication
- Outflow Valve (OFV) Position Indication
- Manifold Pressure Indication
- Flow Line
- Ground Cart



ELEC PAGE

Displays the operation of the electrical busses and contains the following elements:

- IDGs (Integrated Drive Generators)
- AC EXTERNAL POWER (AC GPU)
- APU (AUXILIARY POWER UNIT)
- TRUs (Transformer Rectifier Units)
- BATTERIES
- RAT(RAM AIR TURBINE)



FLIGHT CONTROL PAGE

Displays the control surfaces position status.

- **Note:** the STATUS section is always on ON or STBY in the current version of the simulator
- ELEC PBIT and HYDR PBIT are Digital remaining time readouts which display the hours until the electrical and hydraulic Power On Built In Tests expire.
- PBIT are automatic tests that happen after power-up and take 3 minutes to complete (Electrical PBIT) and 1 minute to complete (Hydraulic Power up test)

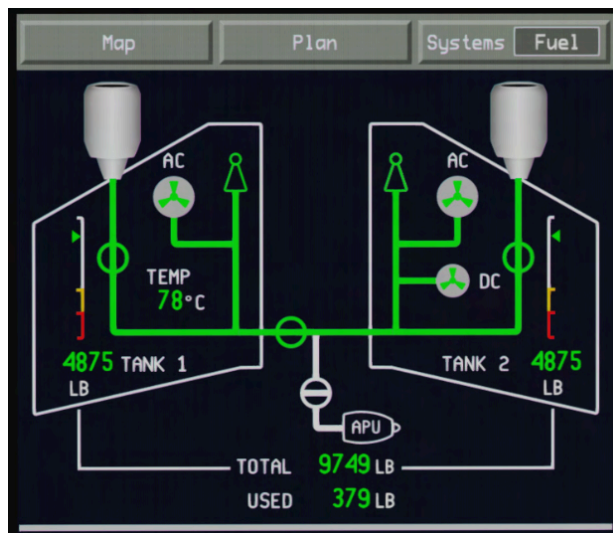


Note: PBIT is not simulated in this version of the product, therefore the readouts change over time but never expire.

FUEL PAGE

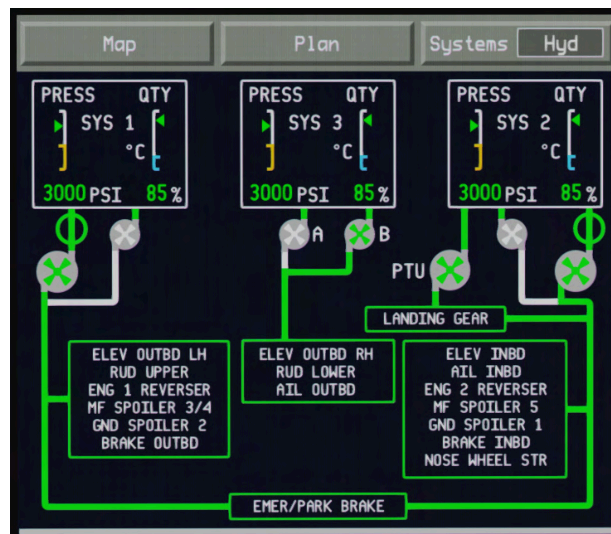
Displays the status of the fuel system on the airplane. The following indications are displayed:

- Temperature indication
- Jet pump indication
- Electric Fuel Pump Status
- Fuel System Valves Status
- Fuel Tank Quantity Indication (Tank 1 and Tank 2)
- Fuel Flow lines
- Total Fuel Quantity indication
- Total Fuel Used indication



HYDRAULIC SYSTEM PAGE

- Pressure indication
- Quantity indication
- Temperature indication
- Engine Pump Shutoff Valve Status
- Electric Hydraulic Pump Status
- Engine Driven Hydraulic Pump Status
- Power Transfer Unit Status
- Hydraulic Flow line
- Hydraulic System Distribution Boxes



INTEGRATED ELECTRONICS STANDBY SYSTEM (IESS)



- An integrated standby instrument system (ISIS) is an electronic aircraft instrument.
- It is intended to serve as backup in case of a failure of the standard glass cockpit instrumentation, allowing pilots to continue to receive key flight-related information.
- Takes about 90 seconds to align.

BRIGHTNESS ADJUSTMENT BUTTONS (+ -)

- Adjusts brightness

AIRSPEED ROLLING DIGITS

- Indicates actual calibrated airspeed.

ILS BUTTON

- Selects the display of ILS 1 deviations.

REFERENCE BAROMETRIC PRESSURE INDICATION

- Indicates the barometric pressure as set by the barometric rotary knob.

STANDARD BUTTON

- Sets the barometric pressure to standard atmospheric pressure.

ALTITUDE INDICATION

- Indicates actual altitude.
- Green hashed box appears on the left side of the numeric digits when altitude is below 10,000 ft.
- Minus (-) symbol appears on the left side when altitude is below sea level.

BARO ROTARY KNOB

- Adjusts the barometric setting.

CAGE BUTTON

- Resets attitude to zero, eliminating accumulated drift when the button is pressed for more than 2 seconds.
- **Note:** This feature is not simulated in this version.

AUTOBRAKE SELECTOR KNOB



- RTO: Selects Rejected Takeoff deceleration rate.
- OFF: Autobrake is deactivated.
- LO: Selects the low deceleration rate.
- MED: Selects the medium deceleration rate.
- HI: Selects the high deceleration rate.

Autobrake Aural warning

The autobrakes will be disengaged automatically with an audible "Autobrake" warning when:

- Pedal braking is applied while wheel speed is above 60 kt in RTO mode and at any speed in landing modes if brake is pressed more than 20% .
- Brake failure is detected while wheel speed is below 60 kt in RTO mode and at any speed in landing modes.
- Either thrust lever is advanced during autobrake application.

PROX TERR INHIB BUTTON & EMERG/PRKG BRAKE LIGHT PANEL



1. PROX TERR INHIB BUTTON

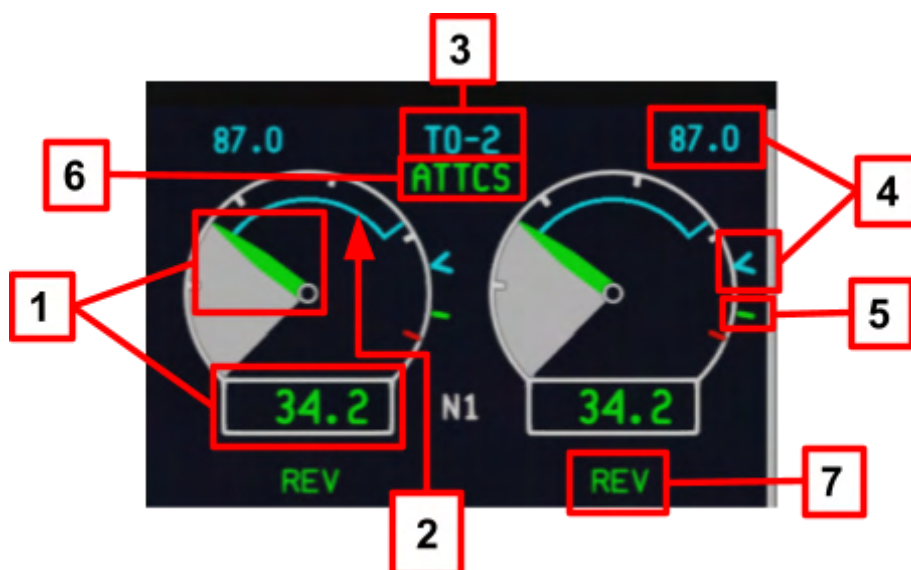
- When pushed out (WHITE STRIPE), inhibits EGPWS and thus avoids unwanted terrain alerts in airports not covered by EGPWS database.
- **Note:** This is currently not simulated

2. EMERG/PRKG BRAKE LIGHT

- ON: Illuminates when the emergency/parking brake is actuated and there is hydraulic pressure on the brake line

EICAS DISPLAY





1. N1 INDICATION

- Displays the percentage of N1 rpm.
 - Green - normal operating range
 - Red - Operating limit exceeded
 - Amber dashed - invalid information or value out of displayable range.
- **OFF** Displays when the engine is shut down.

2. N1 REQUEST INDICATION

- Indicates the momentary difference (transient) between actual N1 and requested N1 applied by TLA.

3. THRUST RATING MODE INDICATION

- Indicates the current thrust-rating mode.

ENGINE THRUST RATINGS

- FADEC calculates the maximum N1 for each thrust rate mode

Available thrust modes:

TAKEOFF (TO-1, TO-2)

- Thrust used for a normal takeoff.
- Manual switching between the takeoff modes is possible in flight on the MCDU-TRS page.

MAXIMUM CONTINUOUS RATING (CON)

- Used in emergency situations where thrust higher than usual is required.

MAXIMUM CLIMB RATING (CLB-1, CLB-2)

- Whenever the selected takeoff thrust is lower than CLB-1 the CLB-2 mode becomes the default.
- Manual switching between the climb modes is possible in flight on the MCDU-TRS page.

MAXIMUM CRUISE RATING (CRZ)

- Maximum N_1 , that can be used for a normal cruise flight.
- Automatically selected when level at the Cruise altitude

4. N1 TARGET INDICATION (DIGITAL READOUT AND BUG)

- Maximum N1 for the engine thrust rating mode indicated on EICAS.

5. MAXIMUM N1 INDICATION

- Indicates the maximum allowable N1 (maximum thrust) for the current thrust rating and operating conditions.

6. ATTCS INDICATION

- Displayed to indicate the Automatic Takeoff Thrust Control System status.
- Green - engaged (only when TL reach TO/GA position), White-armed, Blank - not selected.

AUTOMATIC TAKEOFF THRUST CONTROL SYSTEM (ATTCS)

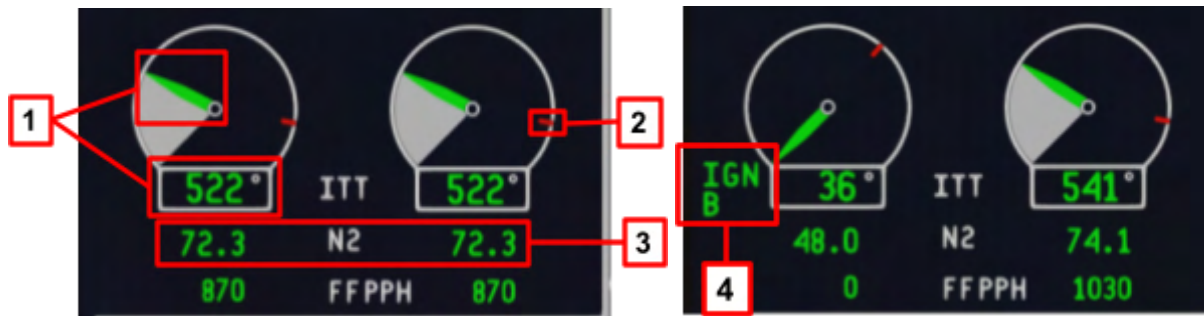
- Controlled by the FADEC and is available on takeoff and go-around.
- Selection is made in the MCDU on the TAKEOFF DATASET MENU
- Takeoff with ATTCS OFF is prohibited.

7. THRUST REVERSER INDICATION

- Indicates the thrust reverser position.
- Green - fully deployed, Amber - in transition
- Red - discrepancy between selected and actual reverser position.

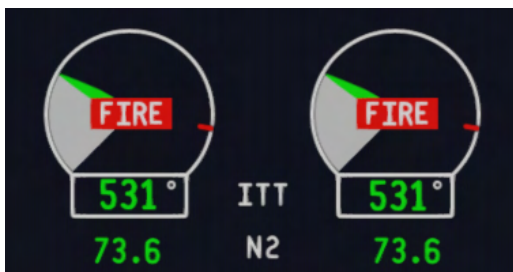
8. N1 RED LINE

- Indicates the N1 limit.



1. INTERTURBINE TEMPERATURE (ITT) INDICATION

- Green - normal operating range, Red - operating limit exceeded.
- FIRE Indicates engine fire condition.



2. ITT RED/AMBER LINE

- Maximum allowable ITT for the current flight phase.
- Limits thrust, thereby avoiding the maximum allowable ITT to be exceeded.

3. N2 INDICATION

- Displays the percentage of N2 rpm.
- Green - normal operating range, Red - operating limit exceeded.

4. IGNITION CHANNEL INDICATION

- Indicates the enabled ignition channel



1. FUEL FLOW INDICATION

- Indicates the left and right engine fuel flow in pounds per hour.

2. FUEL QUANTITY INDICATION

- Indicates left and right wing tanks fuel quantities separately.
- Total fuel in all tanks is indicated inside a gray box.
 - **Green** - normal operating range
 - **Amber** - cautionary operating range
 - **Red** - low fuel quantity.

FUEL TANKS

- Two integral fuel tanks, one in each wing. Each tank incorporates: collector tank and a surge tank.
- Collector tank (inboard section) keeps the fuel pumps submerged, ensuring a constant fuel flow to the respective engine.
- Surge tank (outboard section) is designed to collect any fuel that may enter the vent system during wing down and uncoordinated maneuvers.
- Vented through a vent tank in the surge tank.
- Fuel cannot be transferred from tank to tank or dumped overboard.

FUEL QUANTITY INDICATION

- Electrical sensors measure the fuel quantity in each tank.
- Declare min fuel at 2660 lbs/1200kg.
- Declare emergency fuel at 1300 lbs / 600kg.



1. OIL PRESSURE INDICATION

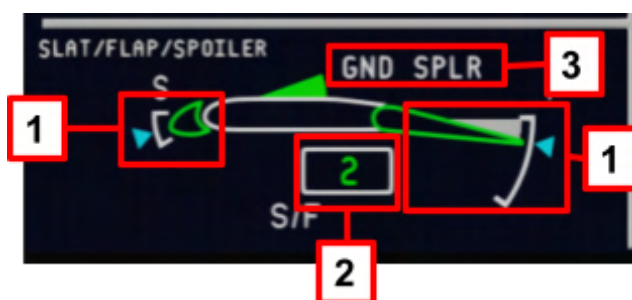
- Indicates the engine oil pressure.
- Green - normal operating range, Amber-cautionary operating range, Red-operating limit exceeded.

2. OIL TEMPERATURE INDICATION

- Indicates the engine oil temperature.
- Green - normal operating range, Amber - cautionary operating range.

3. ENGINE VIBRATION INDICATION

- Indicates low-pressure (LP) and high-pressure (HP) vibration levels for both engines.
- Transient LP or HP high vibration indication without any other abnormal engine parameter indication may be considered normal.



1. SLAT/FLAP POSITION

- Displays the slat/flap position. If the information is invalid, the indication will be removed from the display.
- Cyan pointers show the slat/flap commanded position and the green slat/flap depictions show the actual slat/flap position.

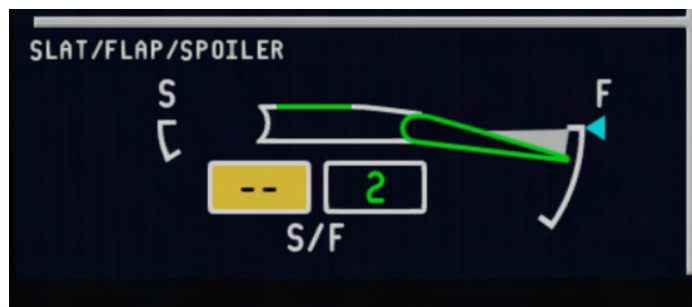
2. SLAT/FLAP READOUT

- Displays the slat/flap surface position.
- Green dashes indicate slat/flap in transit.

3. SPEEDBRAKE INDICATION

- Displays a white SPDBRK annunciation when the airplane is in the air.
- Displays a white GND SPLR annunciation when the airplane is on the ground.

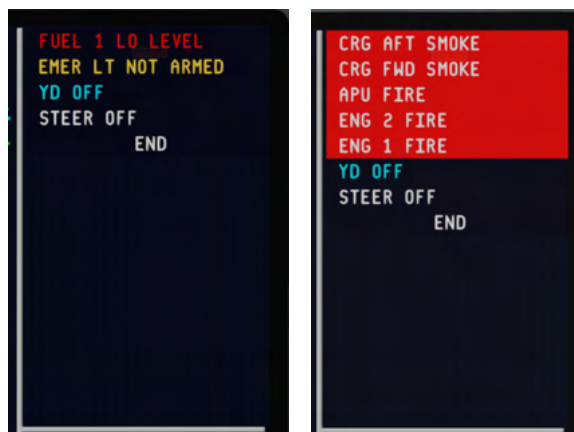
Note: During DC power, not all systems are operational and therefore the SLAT/FLAP/SPEEDBRAKE indication looks like this:



The indications will display normal once AC power has been established.

CREW ALERTING SYSTEM (CAS) WINDOW

Four message priority levels: **Warning, Caution, Advisory, Status**



- **WARNING (red):** Indicates an emergency or airplane system condition that requires immediate corrective or compensatory crew action.
- **CAUTION (amber):** Indicates abnormal operational or airplane system conditions that require immediate crew awareness and should require a subsequent corrective or compensatory action.
- **ADVISORY (cyan):** Indicates operational or airplane conditions that require crew awareness. Subsequent or future crew action may be required.
- **STATUS (white):** Indicates information/status messages.



LANDING GEAR POSITION INDICATION

- **UP:** the white UP label inside a white box indicates that the respective landing gear is up and locked.
- **TRANSIT:** the amber cross-hatched box indicates that the respective landing gear is in transit.
- **DOWN:** the green DN label inside a green circle indicates that the respective landing gear is down and locked.

AUTOBRAKE EICAS INDICATION

- Autobrake status is shown on EICAS in the LG/AUTOBRAKE box.
 - A-BRK indicates the autobrake is armed.
 - Cyan labels indicate the selected mode: **RTO HI MED LO**
 - **---** Indicates the data is invalid

AUXILIARY POWER UNIT (APU)

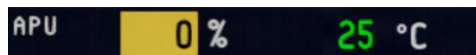
- The APU indication shows OFF if the APU is off.

APU START

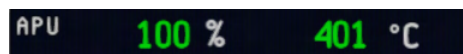
- Rotating the APU switch to the ON position displays yellow dashes for 3 seconds. It is not possible to rotate the APU switch to the START position while the dashes are displayed.



- After 3 seconds, the display changes to a yellow box indicating the APU is ready to be started.



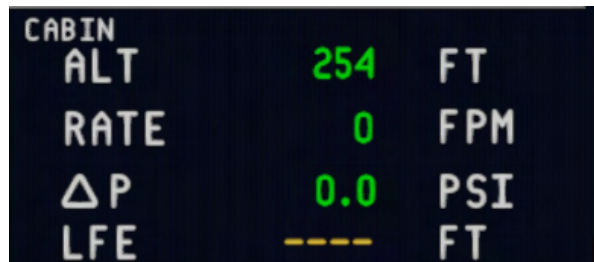
- Rotating the switch to the momentary START position starts the APU.
- It takes about 1 minute for the APU to start.



APU SHUTDOWN

- Rotating the APU switch to the OFF position will initiate the APU cooldown sequence (about 1 minute) followed by the APU shutdown sequence.
- Rotating the APU switch back to the ON position during the shutdown sequence will cancel the APU shutdown.

CABIN ALTITUDE INDICATION

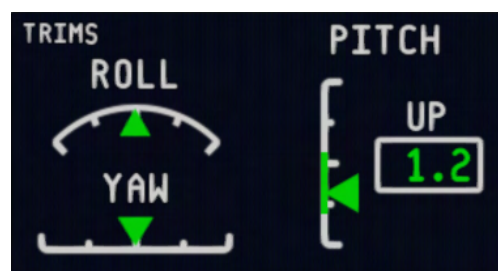


Indicates the following parameters:

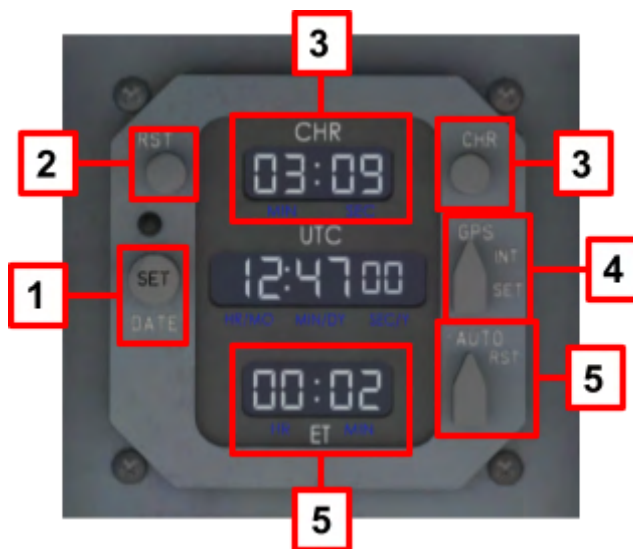
- Cabin altitude indication
- Cabin rate of change indication
- Differential pressure indication
- Landing Field Elevation indication (This is taken from the FMS)

TRIM INDICATION

- Displays the Roll/Pitch/Yaw trim scales
- There is a green takeoff band on the pitch trim scale, corresponding to the allowable pitch trim position for takeoff.
- Pitch Digital trim readout displays the horizontal stabilizer trim position.
UP/DOWN is displayed above or below the readout.
 - UP means Nose up
 - DOWN means Nose down



CLOCK/CHRONOMETER



1. DATE/SET KNOB/BUTTON

- **Note:** This feature is not simulated in the current version. The clock follows the time of the simulator.

2. RESET BUTTON

- Resets the chronometer to zero.
- Display is blanked when the RST button is pressed and the chronometer was stopped, because the value is now 00:00.

3. CHRONOMETER BUTTON

- Starts/stops the chronometer.

4. GPS/INT/SET SELECTOR

- **Note:** This feature is not simulated in the current version. The clock follows the time of the simulator. The selector is permanently on GPS.
- GPS: synchronizes with UTC and DATE from GPS.

5. ELAPSED TIME SELECTOR

- AUTO: automatically starts the chronometer on liftoff. This is always active.
- RST: Resets the elapsed time if Weight-on-Wheels (wow) is present.

LANDING GEAR



LANDING GEAR LEVER

- **UP:** Retracts the landing gear.
- **DOWN:** Extends the landing gear.

DOWNLOCK RELEASE BUTTON

- Lock protects against inadvertent landing gear retraction when weight-on-wheels.
- **Note:** This feature is not simulated in the current version

EMERGENCY LOCATOR TRANSMITTER (ELT) ARTEX ELT

- This feature is not simulated in the current version



GND PROX G/S INHIB BUTTON



- Used to manually cancel glideslope alerts
- Illuminates when pressed any time below 2000ft nominal radar altitude and will be automatically reset (light off) by climbing above 2000ft nominal or descending below 30ft
- **Note:** LG WRN INHIB is not simulated in the current version

ADS PROBES HEATER BUTTON

- **Note:** This feature is not simulated in the current version



PEDESTAL & CONTROL WHEEL

POWERPLANT POWER



1. START/STOP SELECTOR KNOB

- STOP: commands the FADEC to shut down the engine, provided the associated thrust lever is in the IDLE position.
- RUN: normal position for engine operation.
- START (momentary action): initiates the engine start sequence

2. IGNITION SELECTOR KNOB

- OFF: deactivates the ignition system.
- AUTO: FADEC automatically controls the ignition system, depending on engine requirements.
- OVRD: enables FADEC to continuously activate both exciters when the engine is running.
- **Note:** OVRD feature is not simulated in the current version
- **Note:** This knob is by default on the AUTO position and it does not need to be touched during normal startup sequence.

FLIGHT CONTROL MODE BUTTONS



- **Note:** This panel is not simulated in the current version

T/O CONFIG BUTTON

Pressing this button checks the takeoff configuration:

- Flap position based on the FMS takeoff performance pages.
- Parking brake OFF.
- Pitch trim configured for TO.
- Spoiler panels not deployed.



EICAS FULL BUTTON

Automatic de-clutter occurs after landing gear retraction and flap/slat retraction, if all parameters are displaying normal indications. Pressing this button will show the decluttered information.

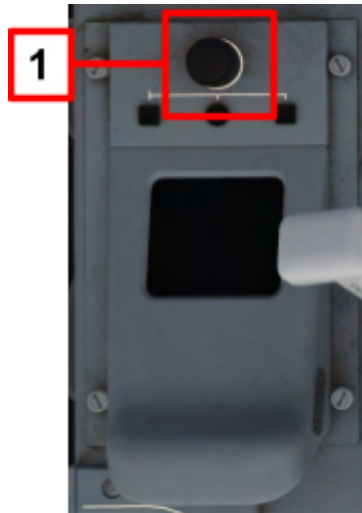
- **PUSH OUT (Dark):** Enables the automatic EICAS de-clutter logic.
- **PUSH IN (White stripe):** EICAS full information presented



De-clutter is disabled when the landing gear is extended and/or flap/slat is not at 0.

CURSOR CONTROL DEVICE (CCD)

In real life, this serves as a mouse that allows the pilots to control the screens. In the simulator, all the functional features on the screens can be directly manipulated with the cursor of your mouse on the screen.



1. The only functionality we left on the CCD was the knob.

- The inner ring of the knob is used for cycling through the flight plan when you are in the PLAN mode on the MFD
- The outer ring of the knob is used for changing the range on the map.
- To make it more user friendly, we have also added a manipulator directly on the map, located on the half range indicator, that can be used to zoom in and out of the map.

STEEP APPROACH SYSTEM

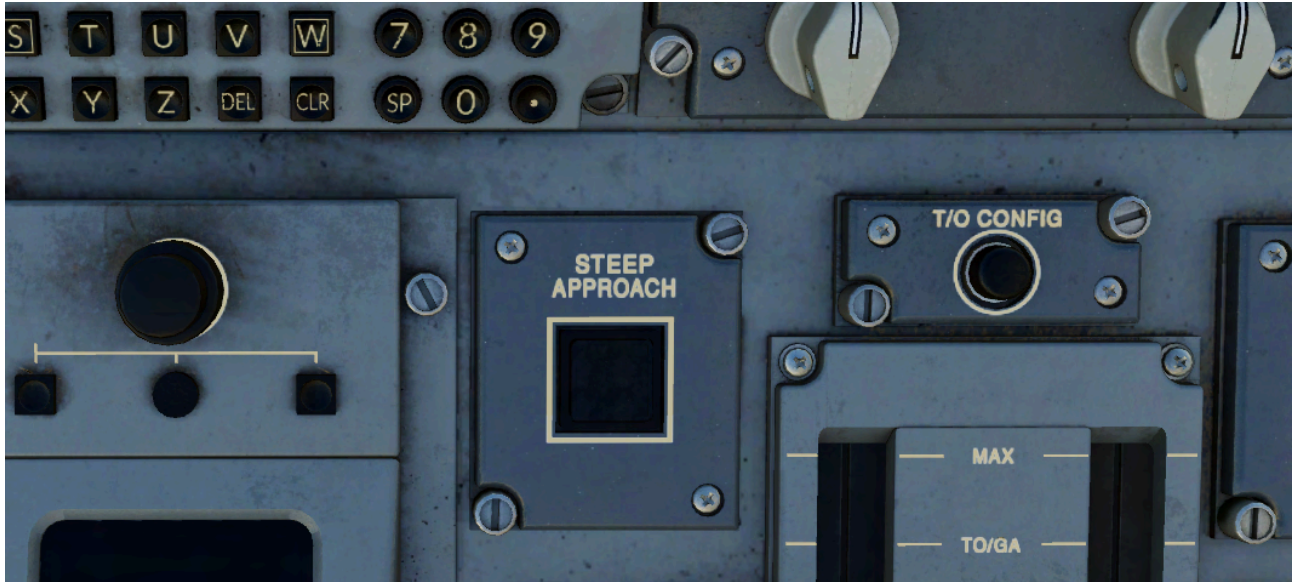


The Steep Approach system, which is only installed on the E170, E190, and Lineage 1000 in real life and in our E-Jets, allows the aircraft to land at airports where the approach angle is greater than 4.5° .

For example, the approach angle at London City Airport (picture above) is 5.5° , nearly twice the standard 3° used in most standard approaches! This makes for very thrilling and challenging approaches.

The steep approach mode dynamically deploys the outer two spoilers during the approach to increase drag and maintain higher N1 values during the approach. This is to minimize the time needed to spool up the engines in case of a Go-Around.

STEEP APPROACH SWITCH



- A steep approach switch in the central pedestal below the left FMS is used to arm/activate the steep approach mode.
- When this switch is pushed, the steep approach mode is armed, as seen by a **STEEP** annunciator in white on the top left of the flightmode annunciator (FMA).
- When conditions are met for the approach, the annunciator is engaged showing **STEEP** in green on the top left of the FMA.
- It automatically disengages when the aircraft touches down, or the go-around mode is activated. The steep approach mode is manually disarmed/disengaged at any time by pushing the switch again.

CONDITIONS TO ACTIVATING STEEP APPROACH

Steep approach is available when all of the following conditions are satisfied:

- There are no failures detected on the following systems:
 - Air Data
 - AHRS
 - Airspeed indicator
 - Electric Gyro system 1
 - Electric Gyro system 2
 - Hydraulic System 1
 - Hydraulic System 2
 - Throttle System
 - Roll Control
 - Pitch Control
 - Steep Approach switch
 - Spoiler Systems
- Pitch is below 7.5°
- AOA is below 14°
- Throttles are below 70%
- Steep approach is enabled.

If there is a failure before the button is pressed the system will indicate error when the button is pushed. The error is indicated with 5 amber flashes on the PFD and SEEP APPR FAIL for 10 secs on the EICAS replaced with persistent STEEP APPR NOT AVAIL. The error can be cleared if the condition that caused it is resolved and the button is pushed to arm the steep approach

Steep approach is armed when all of the following conditions are satisfied:

- Steep approach is available (all conditions above are met)
- The steep approach switch is pushed
- Both throttle lever angles (TLAs) are less than 70
- The AOA is below 10°
- The TOGA button is not pushed or the TOGA button is invalid.
- Gear Down

Steep approach is engaged when all of the following conditions are satisfied:

- Steep approach is armed.
- Flaps set to FULL.
- Three WOW sensors indicate in air.

Steep approach disengages when any of the engagement conditions are no longer satisfied. Once disengaged, steep approach stays disengaged until all of the conditions for arming and engagement are satisfied.

Re-engagement of steep approach requires a pilot action (steep approach toggle switch). This requirement prevents automatic/inadvertent steep approach engagement and spoiler deployment.

Steep Approach Status Indication on the PFD



The status indication for the steep approach mode option is displayed in a single box directly to the left of the approach status indication. The STEEP approach armed, engaged, and disengaged annunciators are located in the STEEP indication window, as shown below.

STEEP APPROACH STATUS ANNUNCIATORS		
Annunciator	Color	Mode Description
STEEP	White	Steep Approach armed
STEEP	Green	Steep approach engaged
STEEP	Amber	Transition from engaged to disengaged

Steep Approach Indicator on the EICAS

The steep approach indicator gives a graphical representation of the position of the steep approach spoilers. When the steep approach spoilers are deployed, the wing surface segment is shown in the up position. When the steep approach spoilers are not deployed, the wing surface segment is shown flat with the upper wing surface. A white STEEP indicator is displayed above the wing when deployed.



POPULAR STEEP APPROACHES AROUND THE WORLD

- **London City Airport (EGLC)**
 - Runways: 09 or 27
 - Angle: 5.5°
- **Lugano Airport (LSZA)**
 - Runway: 01
 - Angle: 6.65°
- **Sion Airport (LSGS)**
 - Runway: 25 (NOT ALIGNED)
 - Angle: Unknown
- **Chambéry Airport (LFLB)**
 - Runway: 18
 - Angle: 4.5°
- **Eagle County Regional Airport (Vail)**
 - Approach: MLS
 - Angle: 8°
- **Innsbruck Airport (LOWI)**
 - Runways: Unknown
 - Angle: 4.0° (Steep Approach not needed as it doesn't meet the 4.5° criteria, can be flown using the E175 or E195)
- **Marseille Provence Airport (LFML)**
 - Runways: Unknown
 - Angle: 4.0° (Steep Approach not needed as it doesn't meet the 4.5° criteria, can be flown using the E175 or E195)
- **Aspen Pitkin County Airport (KASE)**
 - Angle: RNAV 6.49°/ LOC 6.59°
- **Funchal Airport (LPMA)**
 - Runway: 05
 - Angle: 5.0°
- **Gibraltar International Airport (LXGB)**
 - Runway: 09 or 27
 - Angle: 5.0°

SPEEDBRAKE LEVER

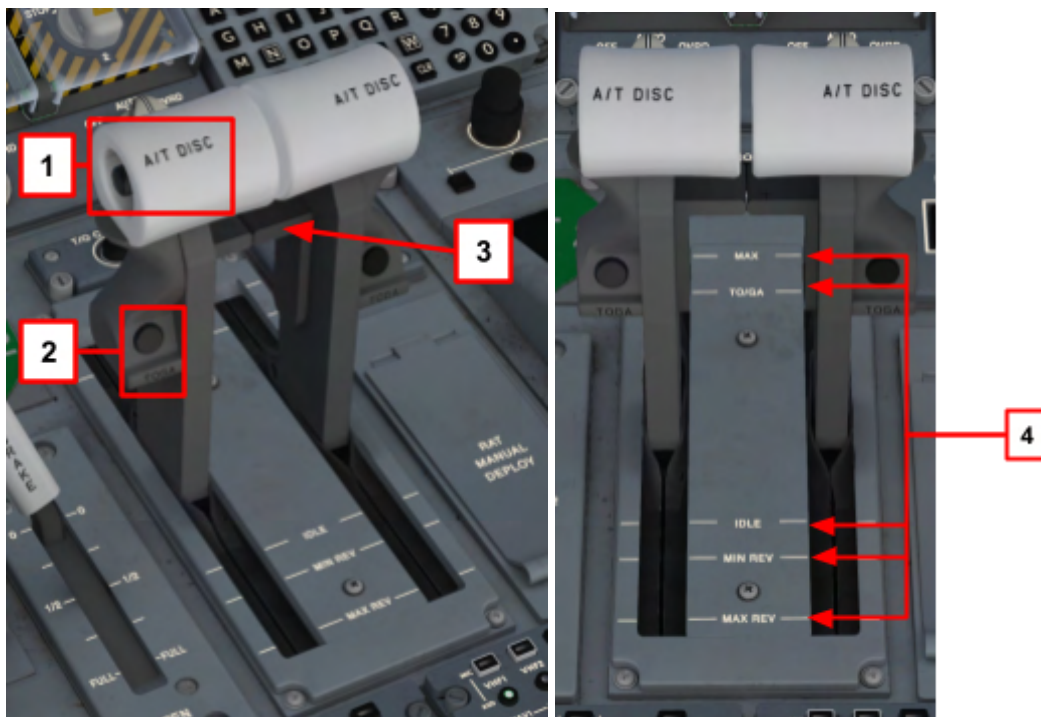


- Symmetrically deploys the multi-function panels. All multi-function spoiler panels deploy at the same angle in response to the speed brake lever position.

SPOILER SYSTEM

- 5 spoiler panels on each wing.
- 3 outboard multifunction spoilers on each wing
 - Multifunction spoilers have 3 modes of operation:
 - **Roll Control:** Deployed asymmetrically for roll augmentation by pilots control wheel.
 - **Speed Brakes:** Deploys all six multifunction spoiler panels symmetrically during flight by speed brake handle position.
 - **Ground Spoilers:** All ten spoiler panels deployed symmetrically to the fully extended position during landing roll.
- Maximum deflection in the air: 30°
- Maximum deflection on the ground: 60°
- Spoilers will not deploy past flaps 1 or below 180 knots

THRUST LEVER QUADRANT



1. AUTOTHROTTLE DISCONNECT BUTTON (A/T DISC)

- Disengages the autothrottles.

2. TAKEOFF AND GO AROUND BUTTON (TO/GA)

- Selects the TO (Take Off) or GA (Go Around) Flight Director Modes.

3. THRUST REVERSER TRIGGER

- Pulling the thrust reverser trigger allows reverser activation when weight on wheels.

4. THRUST LEVER DETENTS

- **MAX:** provides the maximum thrust rating available for dual or single engine operation
- **TO/GA:** selects takeoff, maximum continuous, and go-around mode settings.
- **IDLE:** Selects idle thrust settings.
- **MIN REV:** provides minimum reverse thrust
- **MAX REV:** provides maximum reverse thrust. The thrust lever must be pulled against a spring to achieve the maximum reverse thrust.

RAM AIR TURBINE (RAT)



1. RAM AIR TURBINE (RAT)

- RAT powers the AC ESS BUS
- No altitude restriction for RAT deployment
- RAT provides a lower power output than the APU and IDGs.

2. RAM AIR TURBINE DEPLOYMENT HANDLE

- Manually deploys the Ram Air Turbine which can be used in case of AC power loss in flight.

GND PROX FLAP OVRD BUTTON

- Inhibits triggering flap alerts in case of landings where flap configuration is different from normal landing flap configuration.
- **Note:** This feature is not simulated in this version.



SLAT/FLAP SELECTOR LEVER

- Selects slat/flap position by unlatching the lever and lifting a trigger below the head.



SLATS/FLAP SYSTEM

- Consists of:
 - 4 slats per wing.
 - Two flaps per wing.

Note: Position 4 and 5 are the same slat/flap degrees.

- **Position 4** logic is for **takeoff** and does not provide for the landing gear aural warning on landing.
- **Position 5** logic is for **landing**.

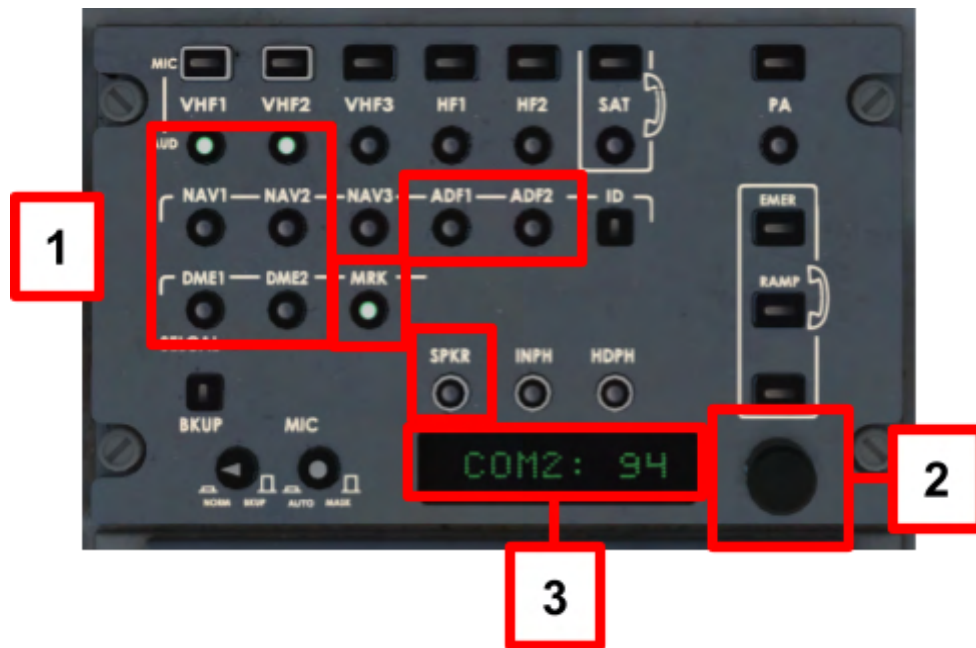
Caution: Using position 4 for landing is prohibited.

- Flap settings approved for takeoff are: 1, 2, and 4.
- Flap settings approved for landing are: 5 and FULL.
- Go-Around:
 - Flaps 5 select flaps 2.
 - Flaps FULL select flaps 4.

AUDIO CONTROL PANEL

1. AUDIO CONTROL BUTTONS

- Pressing the buttons enables reception for selected audio channel.
- The following audio control buttons are functional in the current version, according to the limitations of X-Plane.



2. MASTER VOLUME CONTROL KNOB

- Allows adjustment of the most recently selected audio.

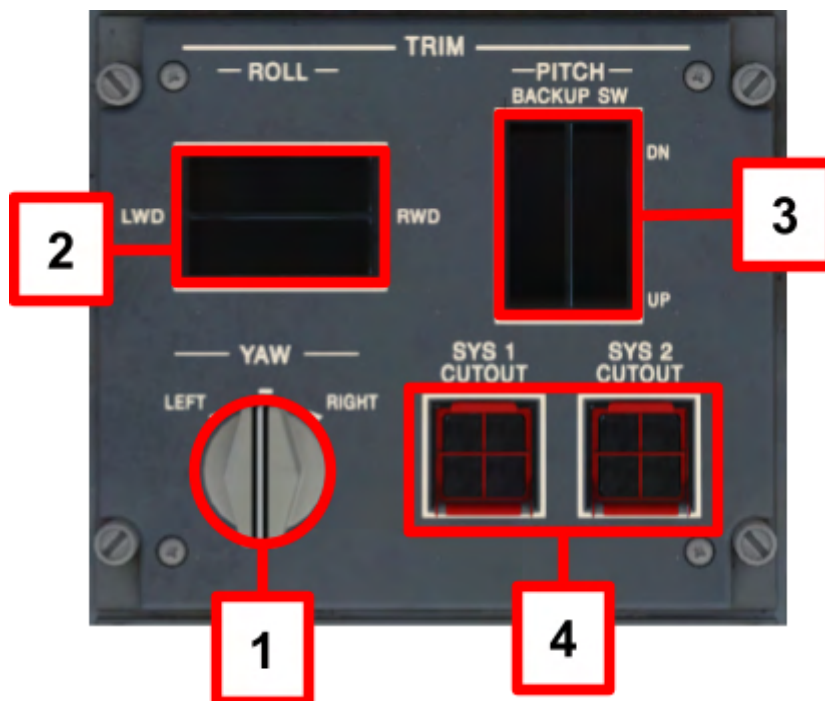
3. ACP DISPLAY

- Displays the selected transmission channel and volume information.

RADIO TUNING

- Radio tuning is performed via the MCDU RADIO page.

TRIM PANEL



1. YAW TRIM KNOB (SPRING-LOADED TO NEUTRAL)

- Actuates the yaw trim to the left or right.

2. ROLL TRIM SWITCH (SPRING-LOADED TO NEUTRAL)

- Actuates the roll trim to the left or right.

3. PITCH TRIM BACK-UP SWITCH (SPRING LOADED TO NEUTRAL)

- Operation of the switch while the autopilot is engaged causes the autopilot to disengage.
- After 3 seconds, an aural warning "TRIM" triggers.

4. PITCH TRIM SYS 1 & SYS 2 CUTOUT BUTTON (GUARDED)

- **Note:** This feature is not simulated in this version

EMERGENCY/PARKING BRAKE HANDLE

- Actuates the emergency/parking brake.
- The handle will lock when pulled up to the fully actuated position.



COCKPIT DOOR CONTROL PANEL

- **Note:** This panel is not simulated in this version.



PASSENGER ADDRESS PTT BUTTON

- **Note:** This panel is not simulated in this version.



ELEVATOR & AILERON DISCONNECT HANDLES

- These handles isolate the CA and FO control columns for elevator / aileron control
- **Note:** This panel is not simulated in this version.



CONTROL WHEEL



HIDING THE CONTROL WHEEL

The yokes can be hidden by:

- Clicking in the middle of them (above the manufacturer's logo)
- By pressing "Y" on your keyboard.



1. PITCH TRIM SWITCH (SPRING-LOADED TO NEUTRAL)

- Trims the airplane when the autopilot is not engaged.

2. CONTROL WHEEL COMMUNICATIONS SWITCH

- **Note:** This feature is not simulated in this version.

3. AP/TRIM DISCONNECT BUTTON

- Pressing once disengages the autopilot.
- Trim-related functions are not simulated

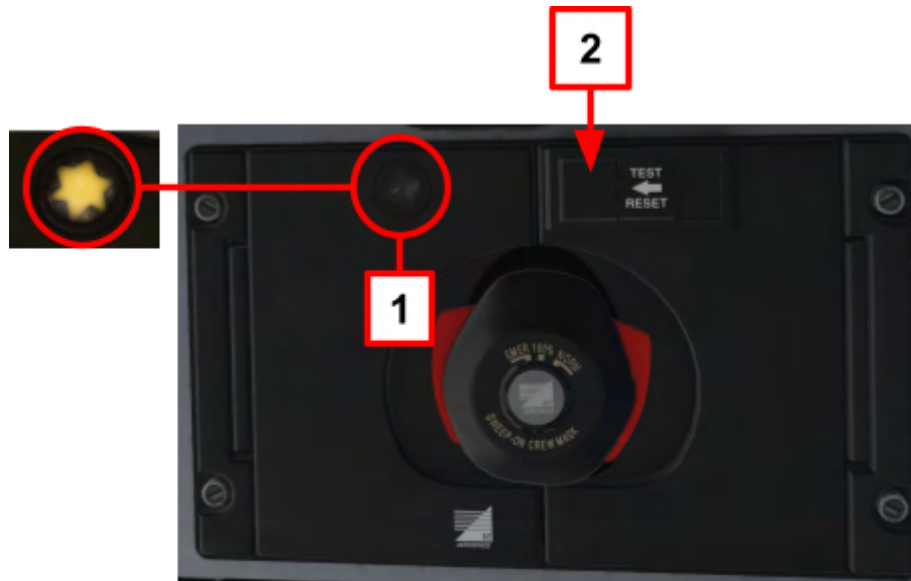
4. TOUCH CONTROL STEERING BUTTON (TCS)

- TCS is currently set to use [X-Plane's Control Wheel Steering mode](#)

5. CHRONOMETER BUTTON

- Starts/stops/resets the chronometer displayed on the PFD

OXYGEN MASK



1. FLOW INDICATOR

- A bright yellow star when visible indicates that oxygen is flowing.

2. TEST/RESET BUTTON (SPRING LOADED)

- Pressing this button with the mask stowed tests the oxygen mask.
- The flow indicator star momentarily blinks and oxygen flow will be audible through the audio system.

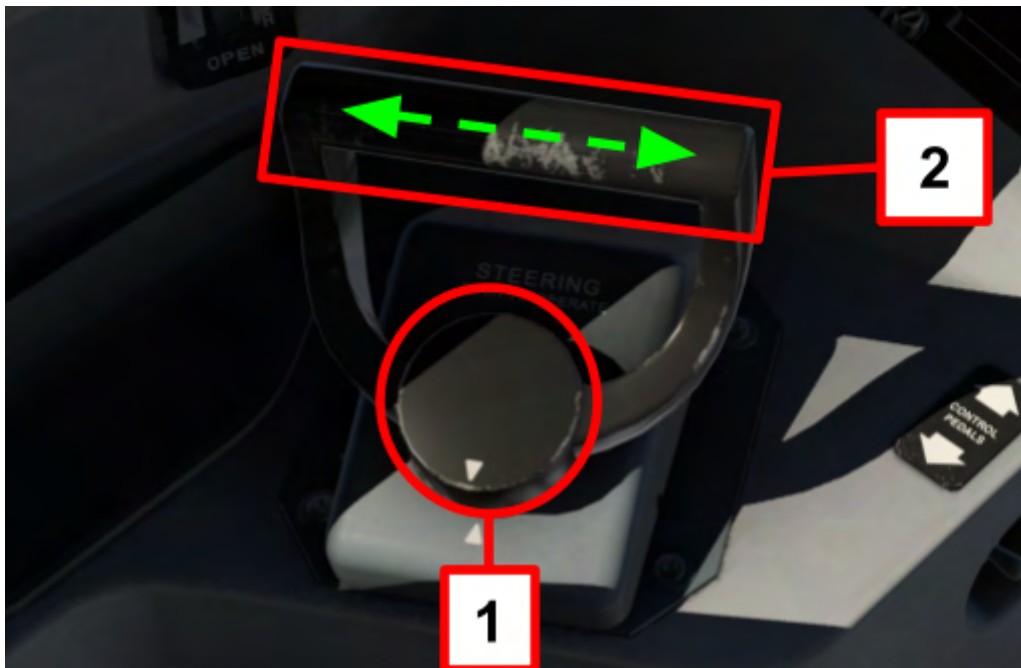
HORN

1. HORN BUTTON

- Pressing this triggers the horn sound



STEERING HANDLE



1. STEERING HANDLE TOGGLE

- The handle must be pushed down to engage the steering system. This is done by clicking on the middle of the handle (1).
- Pressing the handle again will disable steering and will return the handle to the inactive position.

2. STEERING HANDLE MANIPULATOR

- After toggling the steering tiller, the handwheel becomes a manipulator. Rotating the handwheel left or right then commands steering.
- Alternatively, your hardware (mouse, yoke, joystick) can be used for steering.

CUSTOM COMMANDS FOR YOUR HARDWARE

Here are some of the most common custom commands and datarefs which you can use to assign to your hardware. We have hundreds of custom commands, so please be use the search box in the Joystick settings menu to look through the entire list of available custom commands.

Commands: Below you'll see how the commands are named in the X-Plane Settings Menu:

- **AP:** *"Autopilot toggle - also auto controls yaw damper if needed"* (**XCrafts/APYD_Toggle**)
- **AT:** *"Arm/Engage autothrottles (toggle)"* (**XCrafts/ERJ/AutoThrottle**)
- **TOGA:** *"Key_TOGA"* (**XCrafts/ERJ/TOGA**)
- **VNAV:** *"VNAV - Altitude management by the FMS"* (**XCrafts/ERJ/VNAV**)
- **ALT:** *"ALT - Altitude hold"* (**XCrafts/ERJ/alt_hold**)
- **FPA:** *"FPA - Flight Patch Angle altitude change"* (**XCrafts/ERJ/FPA**)
- **FLCH:** *"Altitude change using pitch to maintain speed"* (**XCrafts/ERJ/FLCH**)
- **APP:** *"Select LOC and GS with NAVI"* (**XCrafts/ERJ/APPCH**)
- **Altitude Setting Knob:**
 - *"Altitude down 100"* (**XCrafts/ERJ/ALT_dn_100**)
 - *"Altitude down 1000"* (**XCrafts/ERJ/ALT_dn_100**)
 - *"Altitude up 100"* (**XCrafts/ERJ/ALT_up_100**)
 - *"Altitude up 1000"* (**XCrafts/ERJ/ALT_up_100**)
- **Speed Knob:**

- "Autothrottle speed down 1" (**XCrafts/ERJ/SPD_dn_1**)
- "Autothrottle speed down 10" (**XCrafts/ERJ/SPD_dn_10**)
- "Autothrottle speed up 1" (**XCrafts/ERJ/SPD_up_1**)
- "Autothrottle speed up 10" (**XCrafts/ERJ/SPD_up_10**)

- **VS Knob:**

- "Vertical speed down 10" (**XCrafts/ERJ/VS_dn_10**)
- "Vertical speed down 100" (**XCrafts/ERJ/VS_dn_100**)
- "Vertical speed up 10" (**XCrafts/ERJ/VS_up_10**)
- "Vertical speed up 100" (**XCrafts/ERJ/VS_up_100**)

- **FPA Knob:**

- "FPA angle down 0.1" (**XCrafts/ERJ/FPA_dn_pt_one**)
- "FPA angle down 1.0" (**XCrafts/ERJ/FPA_dn_one**)
- "FPA angle up 0.1" (**XCrafts/ERJ/FPA_up_pt_one**)
- "FPA angle up 1.0" (**XCrafts/ERJ/FPA_up_one**)

Datarefs:

- SPEED: XCrafts/ERJ/autopilot/airspeed_dial_kts_mach
- ALT: XCrafts/ERJ/autopilot/altitude
- VS: XCrafts/ERJ/autopilot/vertical_velocity
- FPA: XCrafts/ERJ/autopilot/fpa

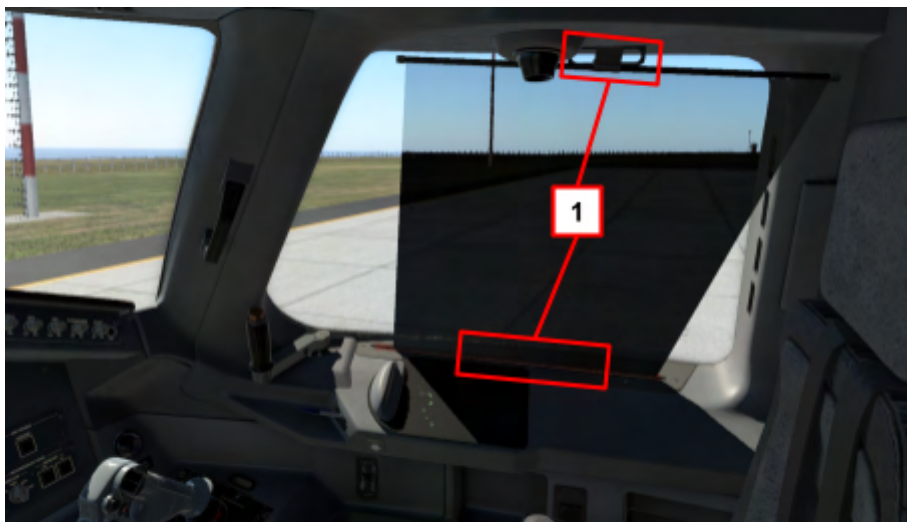
CLICKABLE REGIONS

There are many moving parts around the airplanes. They are all toggled by clicking on them. Here's a list of parts you can manipulate by clicking:

- FWD Main Door (Can also be controlled through the EFB)
- AFT Main Door (Can also be controlled through the EFB)
- FWD Service Door (Can also be controlled through the EFB)
- AFT Service Door (Can also be controlled through the EFB)
- FWD Cargo Door (Can also be controlled through the EFB)
- AFT Cargo Door (Can also be controlled through the EFB)
- Left & Right Cockpit Windows
- Pilot Seats can be moved back and to the side for the pilots to get in the seats
- Pilot Seat armrests can be stowed
- Cockpit door
- Jump Seat
- Spot Lights can be aimed into any allowable direction
- Torches (if you click on the torch holder, you will toggle the flashlight)

I encourage you to visualize the clickable regions through the X-Plane Menu > View and explore the airplane! You may find other moving parts or even some easter eggs ;)

You can also trigger the Side window shades. You'll first need to hide the EFB by clicking on the paper holder, after which you can click on the Window shade (1).



Similarly you can also toggle the front windshield sunshades which are located next to the flashlight, near the pedals. You can then adjust the position of the sunshade on the rail by dragging it.

ENGINE START UP PROCEDURE

We have made a series of videos that will walk you through the engine startup sequence. You can find them on our [website](#) or our [youtube](#) channel.